UNIT – I

PARTICLE TECHNOLOGY

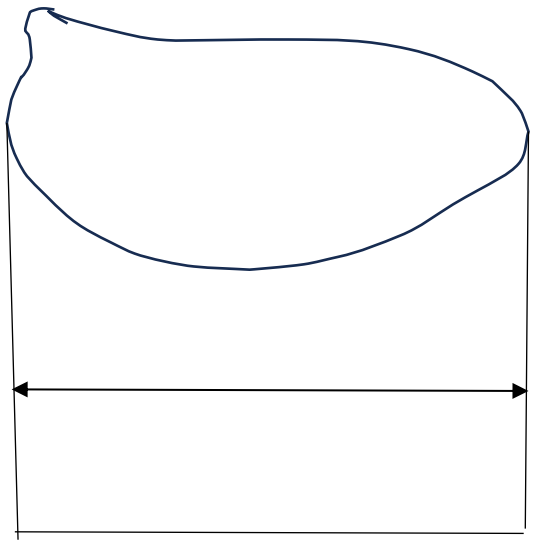
Solids, in general are more difficult to handle than liquids or gases. In processing solids appear in a variety of forms angular pieces, continuous sheets, finely divided powders. They may be hard and abrasive, tough and rubbery, soft or fragile, dusty, cohesive, free flowing or sticky. What ever their forms, means must be found to manipulate the solids as they occur and, if possible, to improve their handling characteristics.

Solids particles are characterized by their size, shape and density. Particles of homogeneous solids have the same density. Where as composite solids such as metal bearing ore have various densities. For regular particles size and shape are easily specified such as sphere and cubes. For irregular particles the terms size and shape are to be defined.

Particle Technology deals with the properties, characteristics, behaviour and applications of particles.

Equivalent Diameter:

The equivalent diameter of a non-spherical particle is sometimes defined as the diameter of a sphere having the same volume as the particle denoted by Dp.
Or
The equivalent diameter of a non-spherical particle is defined as the diameter of a sphere having the same dimension. Same dimensions are sphere of same surface area, sphere of same volume, sphere of same minimum length, sphere passing through the same aperture, sphere of same weight, sphere having same sedimentation rate, sphere having the same settling velocity as the particle, sphere of the same projected area as the particle, as viewed from above, when lying in its position of maximum stability(eg on a microscopic slide) etc
Thus Feret diameter is the distance apart of two parallel planes which just touch the particle.



Ferets Diameter

Sphericity:

The shape of an individual particle is conveniently expressed in terms of the sphericity 'ϕ_s' which is independent of particle size. For a spherical particle $\phi_s = 1$. ϕ_s is defined as the ratio of the surface area of sphere (having equal volume of particle) to the actual surface area of particle.
In another way sphericity is defined as surface to volume ratio of sphere of dia D_P divided by the surface to volume ration for the irregular particle whose equivalent dia is D_P

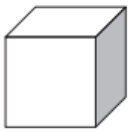
$$\phi_s = \frac{Surface\ area\ of\ sphere\ of\ same\ volume\ as\ particle}{Surface\ area\ of\ particle} = \frac{6V_p}{D_p S_P}$$

Expression for ϕ_s:

$$\emptyset_s = \frac{surface\ to\ volume\ ration\ of\ a\ shpere\ of\ dia\ D_P}{surface\ to\ colume\ ration\ of\ particle\ whose\ nominal\ size\ is\ D_P}$$

$$\therefore \emptyset_s = \frac{\left(\frac{S_P}{V_P}\right)_{sphere\ whose\ volume\ equals\ particle}}{\left(\frac{S_P}{V_P}\right)_{Particle}} = \frac{\frac{\pi D_P^2}{\frac{\pi D_P^3}{6}}}{\frac{S_P}{V_P}} = \frac{6V_p}{D_p S_P}$$

Sphericity of particles having different shapes

Particle name	*iagram*	*Sp ericity* Φ	*Particle name*	*iagram*	*Sp ericity* Φ
Sphere		1.0	Cube		0.81
Sand		0.9	Crushed particles		0.6 to 0.8
Cylinder (length = diameter)		0.87	Flakes		0.2
Tetrahedron		0.671	Octahedron		0.846
Dodecahedron		0.91	Icosahedron		0.939
Ideal Cone		0.794	Hemisphere		0.84

Note: Sphericity is always less than unity because surface of sphere is always less for same volume of any other shape.

Revision: Formulas

Volume of sphere: $\frac{4}{3}\pi r^3$ or $\frac{\pi D^3}{6}$, Surface area of sphere: $4\pi r^2$ or πD^2

Volume of Cube: $a\ x\ a\ x\ a = side\ x\ side\ x\ side = a^3$, Surface area of Cube: $6a^3$

Volume of Cylinder: $\pi r^2 h$, or $\frac{\pi D^2}{4} l$ Surface area of Cylinder: $2\pi rh + 2\pi r^2$ or $\pi D(\frac{D}{2} + h)$
Surface area of Circle: πr^2

Volume of Cuboid: $L\ X\ B\ X\ h$, Surface Area of Cuboid: 2lw+2lh+2hw

Calculate the sphericity of a solid particle of cubical shape (Cubes).

Volume of Cube: a^3, Surface area of cube: $6a^2$
Volume of Sphere: $\frac{4\pi r^3}{3}$, Surface area of Sphere: $4\pi r^2$
Volume of Cube a^3= Volume of Sphere$\frac{4\pi r^3}{3}$ $\therefore r = \sqrt[3]{\frac{3}{4\pi}}a$

$$\phi_s = \frac{Surface\ area\ of\ sphere\ of\ same\ volume\ as\ particle}{Surface\ area\ of\ particle} = \frac{4\pi a^2\left(\sqrt[3]{\frac{3}{4\pi}}\right)^2}{6a^2} = \frac{(4\pi)^{\left(1-\frac{2}{3}\right)}3^{\left(\frac{2}{3}\right)}a^2}{6a^2}$$

$$= \left(\frac{4\pi}{3}\right)^{1/3}\frac{1}{2} = 0.8059$$

Calculate the sphericity for cubes when length equals the diameter ϕ_s = 1.

Proof for above statement:

Cubes - $\phi_s = \frac{6V_p}{S_p D_p} = \frac{6\times a^3}{6a^2\times a} = 1$

Calculate the sphericity of cylindrical particle of 1 cm diameter and length 3 cm

Volume of particle = $\pi r^2 h = \pi\ x\ 0.5^2\ x\ 3 = 0.75\pi$
Volume of sphere= $\frac{4}{3}\pi r_s^3 = 0.75\pi, \therefore r_s = 0.825\ cm$

$$\emptyset_s = \frac{4\pi r^2}{2\pi r(r+h)} = \frac{4\ x\ \pi\ x\ 0.825^2}{2\ x\ \pi\ x(0.5)(3+0.5)} = 0.78$$

Calculate the sphericity of short cylinder whose length equals to diameter:

Reference: https://youtu.be/qO6l8g4rLHc

Volume of cylindrical particle = $\pi \frac{D^2}{4} h$

Volume of sphere= $\pi \frac{D_S^3}{6} = \pi \frac{D^2}{4} h = \pi \frac{D^3}{4}, \quad (D = h)$

, $\therefore D_S = \left(\frac{3}{2}\right)^{1/3} D$

$$\emptyset_S = \frac{\pi D_S^2}{\pi D(\frac{D}{2}+h)} = \frac{\pi\left(\frac{3}{2}\right)^{2/3} D^2}{\pi D(\frac{D}{2}+D)} = \frac{\pi\left(\frac{3}{2}\right)^{2/3} D^2}{\frac{3}{2}\pi D^2} = (2/3)^{1/3} = 0.874$$

Cylinder: How to calculate Equivalent spherical diameter of a Cylinder based on surface area
Ref: (https://youtu.be/99FNo8lrnAc)

Surface Area of Cylinder = Area of Sphere

$$\underset{surface\ area}{\pi D_P^2} + \underset{circularar\ face}{2\frac{\pi D_P^2}{4}} = \pi D_S^2 \therefore D_S^2 = \frac{3}{2} D_P^2$$

By definition

$$\emptyset_S = \phi_s = \frac{Surface\ area\ of\ sphere\ of\ same\ volume\ as\ particle}{Surface\ area\ of\ particle} = \frac{\pi D_S^{2^2}}{\left[\frac{3}{2}\pi D^2\right]} = \frac{\frac{3}{2}\pi D^2}{\left[\frac{3}{2}\pi D^2\right]} = 1$$

Or if take diameter of Cylinder equal to diameter of Sphere

Then By Definition:

$$\phi_s = \frac{Surface\ area\ of\ sphere\ of\ same\ volume\ as\ particle}{Surface\ area\ of\ particle} = \frac{6V_p}{D_p}$$

$$\text{Cylinder } \phi_s = \frac{6\times\left(\frac{\pi D_P^2}{4}\right)\times D_P}{\left[\underset{surface\ area}{\pi D_P^2} + \underset{circularar\ face}{2\frac{\pi D_P^2}{4}}\right] D_P} = \frac{(6/4)\pi D_P^3}{(6/4)\pi D_P^3} = 1$$

For irregular particles ϕ_s is less than unity, for many crushed materials it is between 0.6 or 0.7 (ϕ_s is always less than unity because surface of sphere is always less for same volume of any other shape).

Pb: Calculate the sphericity of Sphere of dia D, Cylinder of dia D, length L and a cube of side D.

$\emptyset_{S\ cylinder} = 0.874$, $\emptyset_{S\ cube} = 0.806$, $\emptyset_{S\ Sphere} = 1$

$$\emptyset_{sphere} > \emptyset_{Cylinder} > \emptyset_{Cube}$$

Shape Factor (a) : Shape factor is defined as Reciprocal of Sphericity
If the shape factor is less than 1, the particle is more elongated and thin.

Particle Size:

Diameter is specified for any equidimensional (proportional dimensions) particles. Particles which are not equidimensional i.e. which are longer in one direction than in others are often characterized by the second longest major dimensions.
Size range of particles (Expressions for size):

Coarse particles – measured in inches / cm.

Fine particles – In terms of screen (mesh) size.

Very fine - Micrometers nanometers

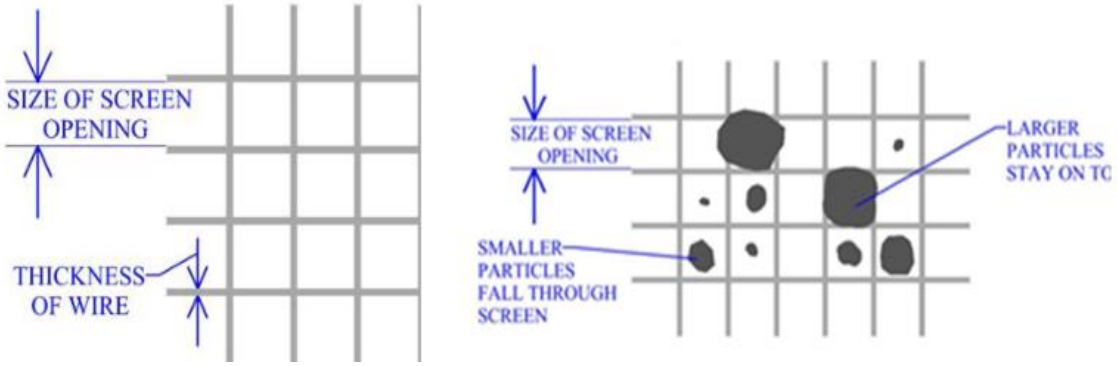
Ultra - In terms of surface area per unit mass.

Standard Screen Series:

(Testing sieves) (measurement of size). Standard screen series are used to measure the size and size distribution of particles in the range between 3, 0.0015. Testing sieves are made up of woven wire screens. The openings are square. Each screen is identified in meshes (openings) per inch the actual openings are smaller than those corresponding to the mesh numbers.

Ex: 40 mesh, 10 mesh

Standard screen series (Testing Seve)
Standardscreen series are used to measure the size and size distribution of particles in the range between 3 and 0.0015 inches. Testing sieves are made up of wire screens. The openings are square. Each screen is identified in meshes (openings) per inch. The actual opening are smaller than those corresponding to the mesh numbers
Eg: 40 mesh, 10 mesh etc



Mesh : Hole of the screen is called mesh

Mesh Number: It is defined as the number of holes per linear inch
Aperture of screen: Aperture is maximum clear space between edges of the screen opening. It is usually given in inches or mm.
Actual and actual opening: Actual opening of the screen is always smaller than the corresponding mesh number because wire diameter is also included.

$$Mesh\ Number = Number\ of\ openings\ per\ linear\ inch$$

$$Clear\ opening = \frac{1}{mesh\ number} - Wire\ thickness$$

$If\ D\ is\ the\ dia\ of\ the\ wire\ or\ wire\ thickness$
$O\ is\ the\ aperture\ \ or\ clear\ openingand$
$m\ is\ the\ mesh\ number, Number\ of\ openings\ per\ linear\ inch$

Then, $O + D = \frac{1}{M}\ in\ inches$ or $O + D = \frac{1}{M}\ x\ 25.4\ in\ mm$

$$\%\ Opening = \frac{O^2}{(O+D)^2}\ x\ 100$$

Some standard screens are:

1) **Tyler Standard Series (TSS),**
2) **British Standard Series(BSS) ,**
3) **Berue of Indian Standards(BIS) ,**
4) **American Society of Testing Machines (ASTM)Standards.**

Tyler standard screen series (Commonly used):

This set of screens is based on the opening of 200 mesh screen which is established at 0.074 mm. The area of opening in any one screen in the screen series is exactly twice that of the openings in the next smaller screen. The ratio of the actual mesh dimension of any screen to that of the next smaller screen is $\sqrt{2} = 1.414$.

$$\frac{A_n}{A_{n+1}} = 2, \qquad \frac{D_n}{D_{n+1}} = \sqrt{2}$$

For closer sizing, intermediate screens are available, each of which has a mesh dimension $4\sqrt{2}$ or 1.189 times that of the next smaller standard screen. Ordinarily these intermediate screen are not used. These sieves are called sub sieves and the analysis carried out using these sieves are called sub sieve analysis.

Pb. Calculate the opening or aperture size in mm of a 200 mesh screen of Tyler standard screen series with a wire of 0.053 mm in diameter. Determine the percentage open area for this mesh size.

$$Clear\ opening = \frac{1}{mesh\ number} - Wire\ thickness$$

Mesh number is number of openings per inch, so it will be 200/inch, 1 inch = 25.4 mm

$$Clear\ opening = \frac{1}{200} - 0.053\ x\ 25.4 = 0.074\ \text{mm}$$

$$\%opening = \frac{0.074^2}{(0.074 + 0.053)^2}\ x\ 100 = 33.95$$

Mixed particle sizes and Size analysis:

In a sample of uniform particles of diameter D_P the total volume of the particles is $\frac{m}{\rho_P}$ where m and ρ_P are the total mass of the sample and the density of the particle, respectively.
If the volume of one particle is v_P, the numbers of particles in the sample 'N' is

$$N = \frac{m}{\int_P V_P}$$

If S_P is the surface area of the particle then the total surface area of the particle

$$A = NS_P$$

Total Surface area , $A = NS_P = \frac{m}{\int_P V_P}. S_P = \frac{mS_P}{\int_P V_P}.$

We know, $\phi_S = \frac{6V_p}{S_p D_p}$ or $\frac{S_P}{V_P} = \frac{6}{\varphi_s D_p}$

$$A = NS_P = \frac{m}{\int_P}. \frac{6}{\varphi_s D_p} = \frac{6m}{\varphi_s \int_p D_p}$$

Specific Surface area of mixture

To apply above equations to mixtures of particles having various sizes and densities the mixture is sorted into fractions each of constant density and approximately constant size [particle size distribution]. Each fraction can then be weighed, individual particles in it can be counted or measured by microscopic methods. The above equations can then be applied to each fraction and the results added to give A_W.

$$A_W = \frac{6\,x_1}{\varphi_s \int_p \overline{D_{P1}}} + \frac{6\,x_2}{\varphi_s \int_p \overline{D_{P2}}} + \frac{6\,x_3}{\varphi_s \int_p \overline{D_{P3}}} + \cdots \ldots \ldots + \frac{6\,x_n}{\varphi_s \int_p \overline{D_{Pn}}}$$

For particles having constant density and sphericity, the specific surface area of the sample is

$$A_W = \frac{6}{\varphi_s \int_p} \sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}}$$

$x_i = mass\ fraction\ in\ a\ given\ increment,$
$n = number\ of\ particles,$
$\overline{D_{Pi}} = Average\ particle\ diameter$
$(Average\ particle\ diameter\ is\ avergae\ of\ smallest\ and\ largest\ particle\ in\ the\ increment)$

Average particle Size

The average particle size of particles is defined in several different ways.

1. **Volume surface mean diameter or *Sauter mean diameter* $\overline{D_s}$** : The most commonly used average particle size is the volume surface mean diameter or *Sauter mean diameter* $\overline{D_s}$ which is related to the specific surface area A_W. It is defined as

$$\overline{D_s} = \frac{6}{\varphi_s A_W \int_p}$$

But $A_W = \frac{6}{\varphi_s \int_p} \sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}}$

$$\therefore \textit{Sauter mean diameter or volume mean diameter } \overline{D_s} = \frac{1}{\sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}}}$$

If the number of particles in each fraction N_i is known instead of the mass fraction, then $\overline{D_s}$ is given by

$$\overline{D_s} = \frac{\sum_{i=1}^{i=n} n_i \overline{D}_{Pi}^3}{\sum_{i=1}^{i=n} n_i \overline{D}_{Pi}^2}$$

$$\overline{D_s} = \frac{1}{\sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}}}$$

Other averages commonly used are

2. **Mass mean diameter $\overline{D_w}$:**

$$\overline{D_w} = \sum_{i=1}^{i=n} x_i \, \overline{D_{Pi}}$$

3. **Volume mean diameter $\overline{D_v}$:**

Dividing the total volume of the sample by the number of particles in the mixture gives the average volume of a particle. This dia is the volume mean diameter $\overline{D_v}$

$$\overline{D_v} = \left(\frac{1}{\sum_{i=1}^{i=n} \frac{x_i}{\overline{D}_{pi}^3}} \right)^{1/3}$$

For particles consisting of uniform particles these average diameters are of course all the same. For mixtures containing particles of various sizes, however the several average diameters may differ widely from another.

4. Number of particles in the mixture:

The numbers of particles in the sample 'N' is

$$N = \frac{m}{\int_P V_P}$$

N_w Total population in one unit mass of sample is obtained by summation overall fractions

For a given particle volume is proportional to its diameter cubed or

$$v_P = aD_p^3$$

Where a is the volume shape factor, and is assumed as independent of size

$$N_W = \frac{1}{a\int_p} \sum_{i=1}^{i=n} \frac{x_i}{\bar{D}_{pi}^3} = \frac{1}{a\int_p \bar{D}_v^3}$$

The specific surface area, the various average diameters, and the number of particles are readily calculated from the particle size analysis through the use of simple computer programs. Many measuring instruments for very fine particles are programmed to report these quantities directly.

Screen Analysis:

The simplest and most common method of separating mixtures by size alone is to make a screen analysis using series of testing series. In making an analysis a set of standard screens is arranged serially in a stack, with the smallest mesh at the bottom & largest at the top. The sample is placed on top of the screen, ie on the top screen and stack is shaken mechanically for a definite period of time. The particles retained on each screen are removed and weighed, and the masses of the individual screen increments are converted to mass fraction or mass percentages of the total sample any particles that pass the finest screen are caught in a pan at the bottom of the stack.

Screen analysis is of 2 types, They are called, Differential screen analysis & Cumulative screen analysis the difference between the two methods of screen analysis is based on the difference in the tabulation or plotting of the results obtained from the screen analysis.

Differential Analysis: Information from particle size analysis is tabulated to show the mass or number fraction in each size increment as a function of average particle size. An analysis tabulated in this way is called Differential analysis.

Since the particles on any one screen are passed by the screen immediately ahead of it, two members are made to specially the size range of an increment, one for the screen through which the fraction posses and the other on which it is retained terms the notation 14 / 20 means "through 14 mesh and 20 mesh" (i.e. passes through 14 mesh but retained on 20 mesh)

Average size for this would be $\frac{14+20}{2}$

-710+300: (-) negative means passing through and (+) positive sign indicates the material is retained on the screen Average size will be $\frac{710+300}{2}$

Tabulation in case of Differential Analysis:

Mesh No	Screen Opening Dpi; mm	Mass fraction retained xi	Avg. particle dia. $\overline{D}_{Pi}$ mm
4	4.699	0.000	-
6	3.327	0.0251	4.013
8	2.362	0.1250	2.845
PAN	-	0.0075	0.037

In the above table , the first two columns give the mesh size and width of screen openings. The third column is the mass fractions of the total sample that is retained in the designated screen. This fraction is x_i, where i is the average opening of varies from 1 to n. The symbol D_{Pi}, means the particle diameter equal to mesh opening of screen i. The last column D_{Pi} is the average diameter of the particles retained on any screen i which is the arithmatic average of the opening of the screen which passes the material and the opening of the mesh which retains the material.

Cumulative Analysis:

A cumulative analysis is obtained from a differential analysis by adding, (cumulatively) the individual differential increments, starting with that retained on the smallest mesh, and tabulating or plotting the cumulative sums against the mesh dimensions of the retaining screen of the last to be added. In the following table the last column gives the cumulative fraction which is smaller than the corresponding value of Dpi. This is called cumulative underflow.

Tabulation ex. (cumulative Analysis):

Mesh. No	Screen opening Dpi, mm	Mass fraction retained xi	Avg. particle dia Dpi mm	Cumulative fraction smaller than Dpi	Cummulative fraction larger than D_{Pi}
4	4.699	0.000	-	1.0000	
6	3.327	0.0250	4.013	0.9749	0.0250
8	2.362	0.1250	2.845	0.8499	0.1250+0.0250
. . .	. . .	. . .	. . .	. . .	
150	0.104	0.0041	0.126	0.106	0.0041+.+.+0.1250+0.0250
200	0.074	0.0031	0.089	0.0075	
PAN	-	0.0075	0.037	0.0000	1.00

The cumulative fraction x_n is defined as follow

$x_n = x_1 + x_2 + x_3 + \cdots + x_n$

i.e. $x_n = \sum_{i=1}^{n} xi$

Note: Sometimes in screen analysis the cumulative fractions are written starting from the top of the stack in this case the last column will be "Cumulative fraction larger than Dpi".
Assumption of Differential analysis :

The assumption made is that all particles in a single fraction are equal is size and that the size is the arithmetical mean of the mesh dimensions of the 2 screens that define the fraction.

Ex: the size of 10 / 14 mesh fraction is $\frac{1.651+1.168}{2}$= 1.410 mm i.d., Dpi = 1.410 mm. The symbol of Dpi is used for such arithmetic average diameter.

In Cumulative analysis, the graph of x_n v$_s$ D_{pi} is treated as a continuous function. In principle, the method based on the cumulative analysis is more precise than that based on the differential analysis because when the cumulative analysis is used, the assumption that all particles in a single fraction are equal in size in not needed. The graph of x_n v$_s$ D_{Pi} is treated as a continuous function.

Problems

Pb 1. Determine the Volume surface mean diameter, Volume mean diameter, specific surface area of a given powder sample using the data given in table. Density of particle is 2650 kg/m3 and sphericity is 0.7

Mesh No.	-	4/6	6/8	8/10	10/14	14/20	20/28	28/35	35/48	48/65	-65
Screen opening Dpi, mm	4.699	3.327	2.362	1.651	1.168	0.833	0.589	0.417	0.295	0.208	0.208
Mass fraction retained	-	0.031	0.103	0.200	0.186	0.152	0.120	0.095	0.065	0.043	0.005

Solution:

$$Volume\ surface\ mean\ Diameter\ \overline{D_s} = \frac{1}{\sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}}}$$

$$Volume\ mean\ diameter\ \overline{D_v} = \left(\frac{1}{\sum_{i=1}^{i=n} \frac{x_i}{\overline{D}_{pi}^3}} \right)^{1/3}$$

Mesh no	Screen opening (D_P	Mass fraction (x_i)	$\overline{D_{Pi}}$	$\frac{x_i}{\overline{D_{Pi}}}$	$\frac{x_i}{\overline{D_{pi}^3}}$
	4.699				
4/6	3.327	0.031	$\frac{3.327+4.699}{2}$=4.013	0.007725	0.00048
6/8	2.362	0.103	$\frac{2.362+3.327}{2}$ =2.8445	0.03621	0.004475
8/10	1.651	0.2	2.0065	0.099676	0.024758
10/12	1.168	0.186	1.4095	0.131962	0.066423
14/20	0.833	0.152	1.0005	0.151924	0.151772
20/28	0.589	0.12	0.711	0.168776	0.333866
28/35	0.417	0.095	0.503	0.188867	0.746483
35/48	0.295	0.065	0.356	0.182584	1.440666
48/65	0.208	0.043	0.2515	0.170974	2.703053
-65	0.208	0.005	0.208	0.024038	0.555623
			$\sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}}$ =1.162737		$\sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{pi}^3}}$ = 6.0275

$$\overline{D_s} = \frac{1}{1.162737} = 0.8604$$

$$\overline{D_v} = \left(\frac{1}{6.0275}\right)^{1/3} = 0.5493$$

Pb 2. The screen analysis shown in the below table applies to crushed quartz. The density of the particle is 2650 kg/m^3 or 0.00265 g/mm^3 And the shape factors are a =2, sphericity 0.571. For the material between 4 mesh and 200 mesh in particle size calculate

(a) A_W *in square millimeter per gram and* N_w *in particles /gm*

(b) $\bar{D}_V$,

(c) $\bar{D}_S$, $(d)\ \bar{D}_w$ $(e)\ N_i$ *for the* $^{150}/_{200}$ *mesh increment*

(f) What fraction of the total numberof particles is in the 150/200 mesh increment

Mesh	Screen Opening	Mass fraction retained
4	4.699	0
6	3.327	0.0251
8	2.362	0.125
10	1.651	0.3207
14	1.168	0.257
20	0.833	0.159
28	0.589	0.0538
35	0.417	0.021
48	0.295	0.0102
65	0.208	0.0077
100	0.147	0.0058
150	0.104	0.0041
200	0.074	0.0031
Pan		0.0075

Solution:

Mesh	Screen Opening	Mass fraction retained	Average diameter in increment $\overline{D_{Pi}}$	Cummulative mass fraction larger than $\overline{D_{Pi}}$	Cummlualtive fraction smaller than $\overline{D_{Pi}}$	$\frac{x_i}{\overline{D_{Pi}}}$	$\frac{x_i}{\overline{D}_{pi}^3}$	$x_i\overline{D_{Pi}}$
4	4.699	0		0	1	0	0	
6	3.327	0.0251	4.013	0.0251	0.9749	0.006255	0.000388	0.100726
8	2.362	0.125	2.8445	0.1501	0.8499	0.043944	0.005431	0.355563
10	1.651	0.3207	2.0065	0.4708	0.5292	0.159831	0.039699	0.643485
14	1.168	0.257	1.4095	0.7278	0.2722	0.182334	0.091778	0.362242
20	0.833	0.159	1.0005	0.8868	0.1132	0.158921	0.158762	0.15908
28	0.589	0.0538	0.711	0.9406	0.0594	0.075668	0.149683	0.038252
35	0.417	0.021	0.503	0.9616	0.0384	0.04175	0.165012	0.010563
48	0.295	0.0102	0.356	0.9718	0.0282	0.028652	0.226074	0.003631
65	0.208	0.0077	0.2515	0.9795	0.0205	0.030616	0.484035	0.001937
100	0.147	0.0058	0.1775	0.9853	0.0147	0.032676	1.037129	0.00103
150	0.104	0.0041	0.1255	0.9894	0.0106	0.032669	2.07421	0.000515
200	0.074	0.0031	0.089	0.9925	0.0075	0.034831	4.397356	0.000276
Pan		0.0075	0.037	1	0			
						$\sum_{i=1}^{i=n}\frac{x_i}{\overline{D_{Pi}}}$ =0.8284	$\sum_{i=1}^{i=n}\frac{x_i}{\overline{D}_{pi}^3}$ = 8.8295	$\sum_{i=1}^{i=n} x_i\,\overline{D_{Pi}}$ = 1.67739

$$A_W = \frac{6}{\varphi_s \int_p}\sum_{i=1}^{i=n}\frac{x_i}{\overline{D_{Pi}}} = \frac{6}{0.571\ x\ 0.00265}\ X\ 0.08284 = 3309\ {mm^2}/{gm}$$

$$N_W = \frac{1}{a\int_p}\sum_{i=1}^{i=n}\frac{x_i}{\overline{D_{pi}^3}} = \frac{1}{2\ x\ 0.00265}x\ 8.8296 = 1679\ {}^{particles}/_{gm}$$

$$\overline{D_v} = \left(\frac{1}{\sum_{i=1}^{i=n}\frac{x_i}{\overline{D_{pi}^3}}}\right)^{1/3} = \frac{1}{8.8296^{1/3}} = 0.4238\ mm$$

$$\overline{D_s} = \frac{1}{\sum_{i=1}^{i=n}\frac{x_i}{\overline{D_{Pi}}}} = \frac{1}{0.8284} = 1.207\ mm$$

$$\overline{D_w} = \sum_{i=1}^{i=n} x_i\,\overline{D_{Pi}} = 1.677\ mm$$

No of particles in the 150/200 mesh increment is found from

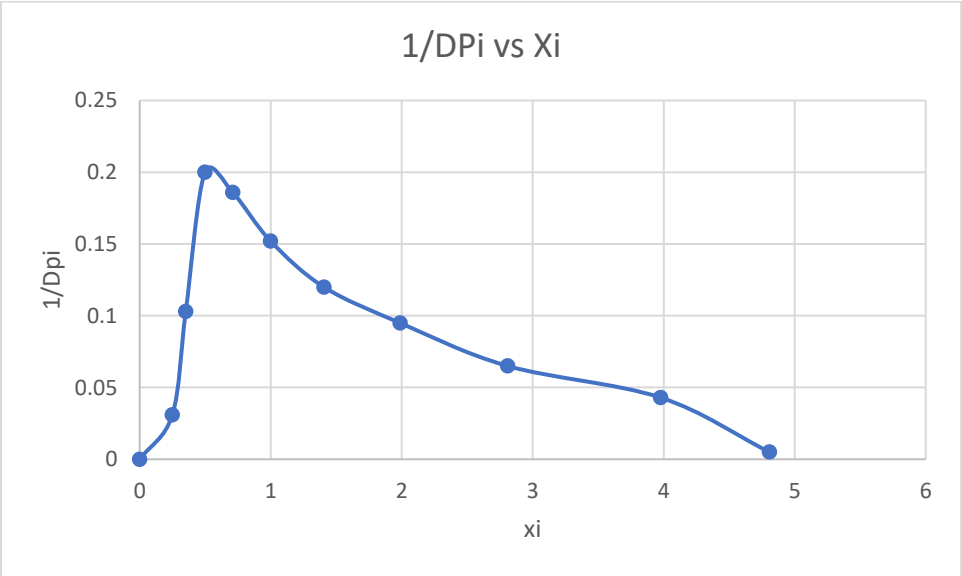
$N_W = \frac{x_2}{a\int_p \overline{D_{p2}^3}} = \frac{0.0031}{2\ x\ 0.00265\ x0.089^3} = 836\ particle/gm,$

$$Total\ \ no\ of\ particles\ N_W\ = 1679\ {}^{particles}/_{gm}$$

This is 836/1679 =0.498 or 49.8 percent of the particles in the top 12 increments.
For the material in the pan fraction, the number of particles and specific surface area are enormously greater than for the coarse material, but they cannot be accurately estimated from the data table.

Volume surface mean dia by Cummulative Analysis:

$$\overline{D_s} = \frac{1}{\int_0^1 \frac{x_i}{\overline{D_{Pi}}}} = \frac{1}{0.55} = 1.8\ mm$$



Pb 3: Calculate the volume surface mean diameter for the following particulate material

Size Range µm	Mass of particles in the range gm
-710+310	30
-300+180	35
-180+90	65
-90+38	70
Pan	55

Solution:

Note: Size range is in micron. And range is between 710 and 310, Therefore the average size is $\frac{710+310}{2} = 505$

Size Range µm	$\overline{D_{Pi}}$	Mass of particles in the range gm	Mass fraction x_i	$\frac{x_i}{\overline{D_{Pi}}}$
-710+310	505	30	(30/255)= 0.117647	0.000233
-300+180	240	35	0.137255	0.000572
-180+90	135	65	0.254902	0.001888
-90+38	64	70	0.27451	0.004289
Pan	38	55	0.215686	0.005676
		Total Weight= 255		$\sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}}$ =0.012658

$$\overline{D_s} = \frac{1}{\sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}}} = \frac{1}{0.012658} = 79.11$$

Pb4: Finely divided clay is used as a catalyst in the petroleum industry. It has a density of 1.2g/cc and a sphericity of 0.5. The size analysis is as follows. Calculate Specific surface area and volume surface mean dia

Average diameter cm)	Mass fraction (g/g)
0.0252	0.088
0.0178	0.178
0.0126	0.293
0.089	0.194
0.038	0.247

Solution:

Average diameter (cm) $\overline{D_{Pi}}$	Mass fraction (g/g) x_i	
0.0252	0.088	3.492
0.0178	0.178	10.000
0.0126	0.293	23.254
0.089	0.194	21.797
0.038	0.247	65.000
		$\sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}} = 123.543$

$$A_W = \frac{6}{\varphi_s \int_p} \sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}} =$$

$$\overline{D_s} = \frac{1}{\sum_{i=1}^{i=n} \frac{x_i}{\overline{D_{Pi}}}} =$$

Mechanical Separations:

Frequently it is necessary to separate the components of a mixture into individual fractions. Procedure for separating the components of mixtures fall into 2 classes.

1. Diffusional Operation (Mass Transfer).

2. Mechanical Separations (M.S)

Mechanical Separations are useful in separating solid particles or liquid drops from fluids.

Different methods of Mechanical Separation are:

1. Screening, 2. Filtration, 3. Sedimentation, 4. Heavy media separation, 5. Differential settling, 6. High tension methods, 7. Magnetic separation.

Mechanical Separations are applicable to **heterogeneous mixtures**. The technique are based on physical differences between the particles such as size and density. They are applicable for separating solids from gases, liquid drops from gases, solids from solids and solids from liquids. The General methods are used for separation.

1. A sieve, or membrane such as a screen or a filter which retains one component and allows the other to pass through.

2. Utilize the difference in the rate of sedimentation of particles as they move through liquid or gas.

3. Special methods – difference in wettability, electrical and magnetic property of substances.

Screening:

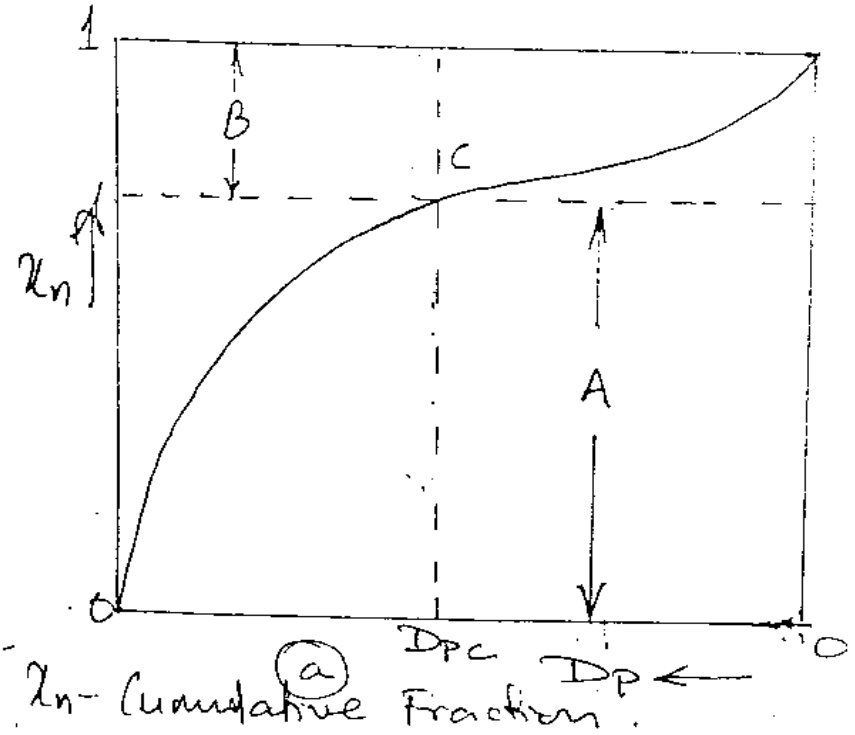
Screening is a method of separating particles according to size alone. In industrial screening the solids are dropped on or through against a screening surface. **The undersize or fine pass through the screen openings. Oversize or tails do not.** A single screen can make but a single separation into two fractions. These are called unsized fractions, because although either the upper or lower limit of the particle sizes they contain, is known, the other limit is unknown material passed through a series of screens of different sizes is separated into sized fractions i.e., fractions in which both maximum and minimum particle sizes are known. Screening is occasionally done wet but much more commonly dry.

Ideal screen or Actual Screens:

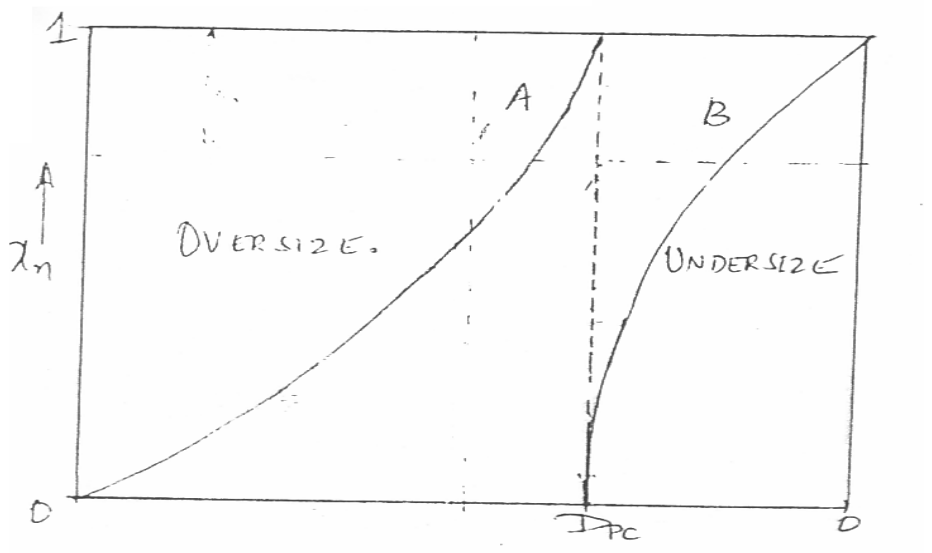
An ideal screen is one which separates the feed mixture in such a way that the smallest particles in the overflow would be just larger than the largest particle in the underflow. Such an ideal separation defines a cut diameter Dpc, which marks the point of separation between the fractions. Dpc is nearly equal to the mesh opening of the screen.

Performance of an ideal screen:

The performance of an ideal screen in terms if the screen analysis of the feed is shown in fig (a). The output is point 'C' in the curve . Fraction A consists of all particles larger than Dpc, and fraction B consists of all particles smaller than Dpc.



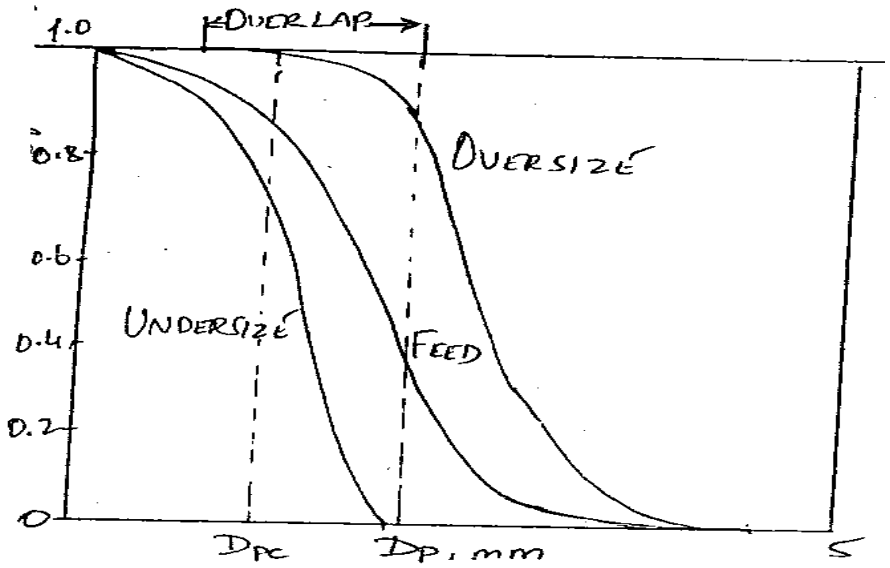
Screen analysis of the fractions A and B are plotted in fig. (b) . The first point on the curve for B has the same abscissa as the last point on the curve for A and there is no overlap of these curves. (Note: In fig (b) oversize and undersize are taken separately and plotted between 0 to 1 and in fig. (a) the entire mixture is taken and plotted between 0 to 1. Also since it is a plot for ideal screen there is a definite cut off point and separates clearly the undersize and the oversize i.e., there is no overlap.



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Actual Screens:

Actual Screens do not give a sharp separation, the overflow has an appreciable content of particles smaller than the designed cut diameter and the underflow has particles larger than the cut diameter. The mass of the 2 leaving streams will not be equal to the individual masses of A and B unless it happens the oversize material in underflow is equal to the undersize in the overflow. The plot of cumulative fraction Vs the particle diameter for actual screens is as shown below.



(Cumulative mass fraction larger than Dpi)

Screen Blindness:

Under the screening action, elongated, sticky, flaky or soft particles may become wedged into the openings of the screens and prevent other particles from passing through. A screen plugged with solid particles is said to be blind.

Screen Effectiveness:

The effectiveness of a screen (often called the screen efficiency) is a measure of the success of a screen in closely separating materials A (desired product) and B (undesired product). A common measure of screen effectiveness is the ratio of oversize material A that is actually in the overflow to the amount of A entering with the feed. Similarly the effectiveness based on material B is defined as the mass of B in the underflow to the amount of B entering with the feed. The relation to find the screen efficiency can be derived doing a material balance over the screen as follows:

Let,
$A = Mass\ flow\ rate\ of\ the\ feed$
$B = Mass\ flow\ rate\ of\ overflow$
$C = Mass\ flow\ rate\ of\ underflow$
$x_A = mass\ fraction\ of$ desired product $in\ feed$
$x_B = mass\ fraction\ of\ desired\ product\ in\ overflow$
$x_C = Mass\ fraction\ of\ desired\ product\ in\ underflow$
$1 - x_A = mass\ fraction\ of\ undesired\ product\ in\ feed$
$1 - x_B = mass\ fraction\ of\ undersired\ product\ \ overflow$

$1 - x_C = Mass\ fraction\ of\ undesired\ product\ in\ underflow$

$$A = B + C \qquad \therefore B = A - C$$

$$Ax_A = Bx_B + Cx_C$$

$$Ax_A = (A - C)x_B + Cx_C$$

$$\frac{C}{A} = \frac{x_A - x_B}{x_C - x_B}$$

$$Ax_A = Bx_B + (A - B)x_C$$

$$Ratio\ of\ oversize\ to\ feed = \frac{B}{A} = \frac{x_A - x_C}{x_B - x_C}$$

$$Effectivensee\ E = Recovery\ x\ Rejection$$

$$Recovery = \frac{Desired\ mateiral\ in\ the\ product}{Desired\ material\ in\ the\ feed} = \frac{B \times x_B}{A \times x_A}$$

$$Recovery = \frac{B \times x_B}{A \times x_A}$$

$If\ Ax_A\ and\ B\ x_B\ are\ desired\ material\ in\ the\ feed\ and\ product\ then$
$A(1 - x_A) and\ B(1 - x_B) are\ the\ undesirable\ material\ in\ the\ feed\ and\ product$

$If\ Cx_C\ is\ the\ desired\ material\ in\ the\ reject,$
$then\ undesired\ material\ in\ reject\ is\ C(1 - x_C)$

$$Rejection = \frac{Undesired\ mateiral\ in\ the\ reject}{Undesired\ material\ in\ the\ feed} = \frac{C(1-x_C)}{A(1-x_A)}$$

$$Undesired\ mateiral\ in\ the\ reject = Undesired\ material\ in\ feed - Undesirable\ material\ in\ product$$

$$C(1 - x_C) = A(1 - x_A) - B(1 - x_B)$$

$$Rejection = \frac{C(1-x_C)}{A(1-x_A)} = \frac{A(1-x_A)-B(1-x_B)}{A(1-x_A)} = 1 - \frac{B(1-x_B)}{A(1-x_A)}$$

$$Effectivensee\ E = Recovery\ x\ Rejection = \frac{B \times x_B}{A \times x_A}\frac{C(1 - x_C)}{A(1 - x_A)} =$$

$$= \frac{B \times x_B}{A \times x_A}\left(1 - \frac{B(1-x_B)}{A(1-x_A)}\right) = \left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\left(1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{(1-x_B)}{(1-x_A)}\right)$$

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right]$$

$$Ratio\ of\ oversize\ to\ feed = \frac{B}{A} = \frac{x_A - x_C}{x_B - x_C}$$

$$Ratio\ of\ underfow\ to\ feed = \frac{C}{A} = \frac{x_A - x_B}{x_C - x_B} = 1 - \frac{B}{A}$$

PB. Powdered coal with following screen analysis is fed to a vibrating 48 mesh screen. The particle size distribution is shown in the table.

(a) Determine the effectiveness of the screen taking oversize as product
(b) Determine the effectiveness of the screen taking under size as product

Mesh Size	%Mass Feed	%Retained Oversize	%Retained undersize
-3+4	1.07	1.8	0
-4+6	2.35	3.3	0
-6+8	6.72	8.8	0
-8+10	8.64	11.2	0
-10+14	10.87	14.2	0
-14+20	17.59	22.9	0
-20+28	13.97	18.2	0
-28+35	10.77	10.4	11.95
-35+48	10.13	6.5	21.98
-48+65	7.46	2.5	23.91
-65+100	5.1	0.2	18.77
-100+150	3.3	0	14.27
-150+200	0.0212	0	0.0912

Solution:
2 methods for solving are shown here:

(1) Without calculating the cumulative mass fraction
(2) With calculating cumulative mass fraction

Method 1: Desired product is oversize and without calculating cumulative mass

Add the fraction retained from 4 mesh to 48 mesh to get x_A, x_B, x_C as it is effectiveness based on oversize

Mesh Number	Mass Fraction Feed	Mass fraction Over size	Mass Fraction under size
4	0.0107	0.018	0
6	0.0235	0.033	0
8	0.0672	0.088	0
10	0.0864	0.112	0
14	0.1087	0.142	0
20	0.1759	0.229	0
28	0.1397	0.182	0

35	0.1077	0.104	0.1195
48	0.1013	0.065	0.2198
65	0.0746	0.025	0.2391
100	0.051	0.002	0.1877
150	0.033	0	0.1427
200	0.0212	0	0.0912
	$x_A = 0.8211$	$x_B = 0.973$	$x_C = 0.3393$

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right]$$

$$= \left[\frac{0.8211 - 0.3393}{0.973 - 0.3393}\right]\frac{0.973}{0.821}\left[1 - \left[\frac{0.8211 - 0.3393}{0.973 - 0.3393}\right]\frac{(1 - 0.973)}{(1 - 0.821)}\right]$$

$$= 0.9\ x\ 0.885 = 0.7975 = 79.56\%$$

Method 2: Desired product is oversize calculating cumulative mass fraction

	Mesh Number	Mass Fraction Feed	Mass Fraction in Oversize	Mass fraction in under size	Cummulative mass Fraction Feed	Cummulative Mass Fraction in Oversize	Cummulative Mass fraction in under size
-3+4	4	0.0107	0.018	0	0.0107	0.018	0
-4+6	6	0.0235	0.033	0	(0.0107+0.0235)= 0.0342	0.051	0
-6+8	8	0.0672	0.088	0	(0.0107+0.0235+0.0672) = 0.1014	0.139	0
-8+10	10	0.0864	0.112	0	0.1878	0.251	0
-10+14	14	0.1087	0.142	0	0.2965	0.393	0
-14+20	20	0.1759	0.229	0	0.4724	0.622	0
-20+28	28	0.1397	0.182	0	0.6121	0.804	0
-28+35	35	0.1077	0.104	0.1195	0.7198	0.908	0.1195
-35+48	48	0.1013	0.065	0.2198	$x_A = 0.8211$	x_B =0.973	x_C =0.3393
-48+65	65	0.0746	0.025	0.2391	0.8957	0.998	0.5784
-65+100	100	0.051	0.002	0.1877	0.9467	1	0.7661
-100+150	150	0.033	0	0.1427	0.9797	1	0.9088
-150+200	200	0.0212	0	0.0912	1.0009	1	1

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right]$$

$$= \left[\frac{0.8211 - 0.3393}{0.973 - 0.3393}\right]\frac{0.973}{0.821}\left[1 - \left[\frac{0.8211 - 0.3393}{0.973 - 0.3393}\right]\frac{(1 - 0.973)}{(1 - 0.821)}\right]$$

$$= 0.9\ x\ 0.885 = 0.7975 = 79.56\%$$

$$Ratio\ of\ overflow\ to\ feed, \frac{B}{A} = \frac{x_A - x_C}{x_B - x_C} = \frac{0.8211 - 0.3393}{0.973 - 0.3393} =$$

$$Ratio\ of\ underflow\ to\ feed, \frac{C}{A} = 1 - \frac{B}{A} = 1 - \frac{0.8211 - 0.3393}{0.973 - 0.3393} =$$

Determine the effectiveness of the screen taking under size as product Without calculating cumulative mass fraction.

Mesh Number	Mass Fraction Feed	Mass Fraction in Oversize	Mass fraction in under size
4	0.0107	0.018	0
6	0.0235	0.033	0
8	0.0672	0.088	0
10	0.0864	0.112	0
14	0.1087	0.142	0
20	0.1759	0.229	0
28	0.1397	0.182	0
35	0.1077	0.104	0.1195
48	0.1013	0.065	0.2198
65	0.0746	0.025	0.2391
100	0.051	0.002	0.1877
150	0.033	0	0.1427
200	0.0212	0	0.0912
	x_A =0.1798	x_B =0.027	x_C = 0.6607

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right] = 0.8853\ x\ 0.9 = 79.76\%$$

Considering cumulative fraction if undersize is desired product: (Cummulative fraction from Top)

Mesh Number	Mass Fraction Feed	Mass Fraction in Oversize	Mass fraction in under size	Cummulative mass Fraction Feed	Cummulative Mass Fraction in Oversize	Cummulative Mass fraction in under size
4	0.0107	0.018	0	0.0107	0.018	0
6	0.0235	0.033	0	0.0342	0.051	0
8	0.0672	0.088	0	0.1014	0.139	0
10	0.0864	0.112	0	0.1878	0.251	0
14	0.1087	0.142	0	0.2965	0.393	0
20	0.1759	0.229	0	0.4724	0.622	0

28	0.1397	0.182	0	0.6121	0.804	0
35	0.1077	0.104	0.1195	0.7198	0.908	0.1195
48	0.1013	0.065	0.2198	0.8211	0.973	0.3393
65	0.0746	0.025	0.2391	0.8957	0.998	0.5784
100	0.051	0.002	0.1877	0.9467	1	0.7661
150	0.033	0	0.1427	0.9797	1	0.9088
200	0.0212	0	0.0912	1.0009	1	1
				$x_A = 1 - 0.8211 = 0.1789$	$x_B = 1 - 0.973 = 0.027$	$x_c = 1 - 0.3393 = 0.6607$

$x_A = 1 - 0.8211 = 0.1789$
$x_B = 1 - 0.973 = 0.027$
$x_c = 1 - 0.3393 = 0.6607$

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right] = 0.8853\ x\ 0.9 = 79.76\%$$

Note: Effectiveness of screen remains same either calculated by oversize as desired product or under size as desired product

Ratio of Product to feed:

$$\frac{B}{A} = \frac{x_A - x_C}{x_B - x_C} = —$$

$$\frac{C}{A} = \frac{x_A - x_B}{x_C - x_B} = 1 - \frac{B}{A}$$

Considering cumulative fraction from bottom

Note: Negative mesh numbers are to be considered if cumulative fraction is considered from bottom

	Mesh Number	Mass Fraction Feed	Mass Fraction in Oversize	Mass fraction in under size	Cummulative mass Fraction Feed	Cummulative Mass Fraction in Oversize	Cummulative Mass fraction in under size
	3				1	1	1
-3+4	4	0.0107	0.018	0	0.9893	0.982	1
-4+6	6	0.0235	0.033	0	0.9658	0.949	1
-6+8	8	0.0672	0.088	0	0.8986	0.861	1
-8+10	10	0.0864	0.112	0	0.8122	0.749	1
-10+14	14	0.1087	0.142	0	0.7035	0.607	1
-14+20	20	0.1759	0.229	0	0.5276	0.378	1
-20+28	28	0.1397	0.182	0	0.3879	0.196	1
-28+35	35	0.1077	0.104	0.1195	0.2802	0.092	0.8805

-35+48	48	0.1013	0.065	0.2198	0.1789	0.027	0.6607
-48+65	65	0.0746	0.025	0.2391	0.1043	0.002	0.4216
-65+100	100	0.051	0.002	0.1877	0.0533	0	0.2339
-100+150	150	0.033	0	0.1427	0.0203	0	0.0912
-150+200							

$x_A = 0.1789$
$x_B = 0.027$
$x_c = 0.6607$

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right] = 0.8853\ x\ 0.9 = 79.76\%$$

Pb. Following is the Particle Size Distribution (PSD) of three cuts obtained from a double deck vibrating screen (48 and 65 mesh). The mass ratio of oversize:Internediate:Undersize is 3:3:4.

(a) Construct particle size distribution for 48 and 65 mesh.
(b) Calculate effectiveness of 48 and 65 mesh screens individually if oversize is desired product and if under size is desired product.
(c) Calculate the effectiveness of double deck vibrating screen system as a whole taking intermediate as product.

	Mass Fraction Retained		
Mesh No	Oversize	Intermediate	Undersize
14	0.0012	0	0
20	0.0068	0	0
28	0.189	0.2	0
35	0.389	0.039	0.001
48	0.337	0.322	0.003
65	0.066	0.526	0.344
100	0.005	0.067	0.299
150	0.005	0.024	0.237
200	0.001	0.002	0.116

Solution:

Let us understand what is a double deck screen here. The sieve set is arranged such that 48 is on the top with lid then will be 65 mesh and below which will be Pan. The sample is placed on 48 mesh and this sieve set is sieved for 15 minutes to get overize retained on 48 mesh, intermediate on 65 mesh and undersize in Pan.

1. Oversize retained on 48 mesh. This is then subjected to sieve Analysis in the same sieve set 14 to 200 to get thevalues in the column oversize mass fraction
2. Intermediates retained on 65 mesh This is then subjected to sieve Analysis in the same sieve set 14 to 200 to get thevalues in the column inermediate mass fraction
3. Undersize in pan, This is then subjected to sieve Analysis in the same sieve set 14 to 200 to get the values in the column undersize mass fraction

(a) Construct particle size distribution for 48 and 65 mesh.

(i) Construct particle size distribution for 48.

Feed to 48 Mesh: To make the PSD, we need to have the feed analysis x_A. Let us find the total material retained on 48 mesh (Given ratio of Oversize:Intermediate:Undersize is 3:3:6 Therefore

Total material retained= 3+3+4=10

Calculations to find $x_{A\ 48\ mesh}$ is shown for 14 mesh

To find $x_A\ for\ 14\ mesh,\quad x_A = \frac{(3\ X\ 0.0012+3\ x0+4x0)}{10} = 0.0036$

Repeat the same for other meshes

(ii) Construct particle size distribution for 65 mesh.

Feed to 65 Mesh: Total material will be 0+3+4=7

(Note: Since material retained on 65 is intermediates we should consider only intermediates and undersize. And we should not consider the oversize)

Calculations to find $x_{A\ 65\ mesh}$ are shown for 14 and 35 mesh. Same can be repeated for other meshes

$\frac{(3x0+4x0)}{7} = 0\ \&\ For\ 35\ mesh, \frac{(3\ x0.039+4x0.001)}{7} = 0.017286$

	Mass Fraction Retained				
Mesh No	Oversize	Intermediate	Undersize	Feed to 48 Mesh	Feed to 65 Mesh
14	0.0012	0	0	0.00036	0
20	0.0068	0	0	0.00204	0
28	0.189	0.2	0	0.1167	0.085714
35	0.389	0.039	0.001	0.1288	0.017286
48	0.337	0.322	0.003	0.1989	0.139714
65	0.066	0.526	0.344	0.3152	0.422
100	0.005	0.067	0.299	0.1412	0.199571
150	0.005	0.024	0.237	0.1035	0.145714
200	0.001	0.002	0.116	0.0473	0.067143

(B)Calculate effectiveness of 48 and 65 mesh screens individually

Effectiveness of 48 mesh if oversize is desired product

	Mass Fraction Retained				
Mesh No	Oversize	Intermediate	Undersize	Feed to 48 Mesh	Feed to 65 Mesh
14	0.0012	0	0	0.00036	0
20	0.0068	0	0	0.00204	0

28	0.189	0.02	0	0.0627	0.008571
35	0.389	0.039	0.001	0.1288	0.017286
48	0.337	0.322	0.003	0.1989	0.139714
65	0.066	0.526	0.344	0.3152	0.422
100	0.005	0.067	0.299	0.1412	0.199571
150	0.005	0.024	0.237	0.1035	0.145714
200	0.001	0.002	0.116	0.0473	0.067143
	x_B=0.923			x_A =0.39288	x_C =0.16557

Add fractions from 14 mesh to 48 mesh to get a cumulative sum.
x_A =0.39288
x_B=0.923,
x_C =0.16557

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right] = 67.8\%$$

Effectiveness of 48 mesh if undersize is desired product

	Mass Fraction Retained				
Mesh No	Oversize	Intermediate	Undersize	Feed to 48 Mesh	Feed to 65 Mesh
14	0.0012	0	0	0.00036	0
20	0.0068	0	0	0.00204	0
28	0.189	0.02	0	0.0627	0.008571
35	0.389	0.039	0.001	0.1288	0.017286
48	0.337	0.322	0.003	0.1989	0.139714
65	0.066	0.526	0.344	0.3152	0.422
100	0.005	0.067	0.299	0.1412	0.199571
150	0.005	0.024	0.237	0.1035	0.145714
200	0.001	0.002	0.116	0.0473	0.067143
	x_C =0.077			x_A =0.6072	x_B =0.8344

Add fractions from 65 mesh to 200 mesh to get x_A =0.6072, x_B =0.8344, x_C =0.077

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right] = 67.8\%$$

Effectiveness of 65 mesh if Over size is desired product

	Mass Fraction Retained				
Mesh No	Oversize	Intermediate	Undersize	Feed to 48 Mesh	Feed to 65 Mesh
14	0.0012	0	0	0.00036	0
20	0.0068	0	0	0.00204	0
28	0.189	0.02	0	0.0627	0.008571
35	0.389	0.039	0.001	0.1288	0.017286

48	0.337	0.322	0.003	0.1989	0.139714
65	0.066	0.526	0.344	0.3152	0.422
100	0.005	0.067	0.299	0.1412	0.199571
150	0.005	0.024	0.237	0.1035	0.145714
200	0.001	0.002	0.116	0.0473	0.067143
		$x_B = 0.907$	$x_C = 0.348$		$x_A = 0.5875$

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right] = 59.76\%$$

Effectiveness of 65 mesh if under size is desired product

	Mass Fraction Retained				
Mesh No	Oversize	Intermediate	Undersize	Feed to 48 Mesh	Feed to 65 Mesh
14	0.0012	0	0	0.00036	0
20	0.0068	0	0	0.00204	0
28	0.189	0.02	0	0.0627	0.008571
35	0.389	0.039	0.001	0.1288	0.017286
48	0.337	0.322	0.003	0.1989	0.139714
65	0.066	0.526	0.344	0.3152	0.422
100	0.005	0.067	0.299	0.1412	0.199571
150	0.005	0.024	0.237	0.1035	0.145714
200	0.001	0.002	0.116	0.0473	0.067143
		x_C =0.093	x_B =0.652		$x_A = 0.412$

$$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right] = 59.76\%$$

Pb. It is desired to separate a mixture of solids into three fraction, coarse fraction retained on 8 mesh, middle fraction retained on 10 mesh and fine fraction in pan. Based on the screen analysis below determine the weight ratio of each of three fractions and the overall effectiveness of 8 and 10 mesh.

Screen	Feed	Oversize	Intermediate	Undersize
3/4	3.5	14	0	0
4/6	15	50	4.2	0
6/8	27.5	24	35.8	0
8/10	23.5	8	30.8	20
10/14	16	4	18.3	26.7
14/20	9.1	0	10.2	20.2
20/28	3.4	0	0.7	19.6
28/35	1.3	0	0	8.9
35/48	0.7	0	0	4.6

Solution:

For 8th mesh if Oversize is desired product:

Method 1: Find mass fraction for the above data and add the mass fraction upto 8 mesh to get x_A x_B x_c

	Mass Fraction			
Screen	Feed	Oversize	Intermediate	Undersize
4	0.035	0.14	0	0
6	0.15	0.5	0.042	0
8	0.275	0.24	0.358	0
10	0.235	0.08	0.308	0.2
14	0.16	0.04	0.183	0.267
20	0.091	0	0.102	0.202
28	0.034	0	0.007	0.196
35	0.013	0	0	0.089
48	0.007	0	0	0.046
	$x_A = 0.46$	$x_B = 0.46$	$x_c = 0.46$	0

Method 2: Take cumulative mass fraction to get x_A x_B x_c

	Mass Fraction				Cummulative mass fraction			
	Feed	Oversize	Intermediate	Undersize	Feed	Oversize	Intermediate	Undersize
4	0.035	0.14	0	0	0.035	0.14	0	0
6	0.15	0.5	0.042	0	0.185	0.64	0.042	0
8	0.275	0.24	0.358	0	0.46	0.88	0.4	0
10	0.235	0.08	0.308	0.2	0.695	0.96	0.708	0.2
14	0.16	0.04	0.183	0.267	0.855	1	0.891	0.467
20	0.091	0	0.102	0.202	0.946	1	0.993	0.669
28	0.034	0	0.007	0.196	0.98	1	1	0.865
35	0.013	0	0	0.089	0.993	1	1	0.954
48	0.007	0	0	0.046	1	1	1	1

For 8th Mesh, considering oversize as desired product:

$$x_A = 0.46, \quad x_B = 0.88, \quad x_C = 0.4 \therefore E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right] =$$

Ratio of Product to feed (8 mesh):

$$\frac{B}{A} = \frac{x_A - x_C}{x_B - x_C} = —$$

$$\frac{C}{A} = \frac{x_A - x_B}{x_C - x_B} = 1 - \frac{B}{A}$$

For 10 mesh, If oversize is the desired product

	Mass Fraction				Cummulative mass fraction			
Screen	Feed	Oversize	Intermediate	Undersize	Feed	Oversize	Intermediate	Undersize
4	0.035	0.14	0	0	0.035	0.14	0	0
6	0.15	0.5	0.042	0	0.185	0.64	0.042	0

8	0.275	0.24	0.358	0	0.46	0.88	0.4	0
10	0.235	0.08	0.308	0.2	0.695	0.96	0.708	0.2
14	0.16	0.04	0.183	0.267	0.855	1	0.891	0.467
20	0.091	0	0.102	0.202	0.946	1	0.993	0.669
28	0.034	0	0.007	0.196	0.98	1	1	0.865
35	0.013	0	0	0.089	0.993	1	1	0.954
48	0.007	0	0	0.046	1	1	1	1
	x_A =0.695	x_B =0.96	x_C =0.708					

$x_A = 0.695, \quad x_B = 0.96, \quad x_C = 0.708 \quad and$

$E = \left[\left(\frac{x_A - x_C}{x_B - x_C}\right)\frac{x_B}{x_A}\right]\left[1 - \left(\frac{x_A - x_C}{x_B - x_C}\right)\left(\frac{1 - x_B}{1 - x_A}\right)\right]$=33.5

Ratio of Product to feed(10 mesh):

$$\frac{B}{A} = \frac{x_A - x_C}{x_B - x_C} = —$$

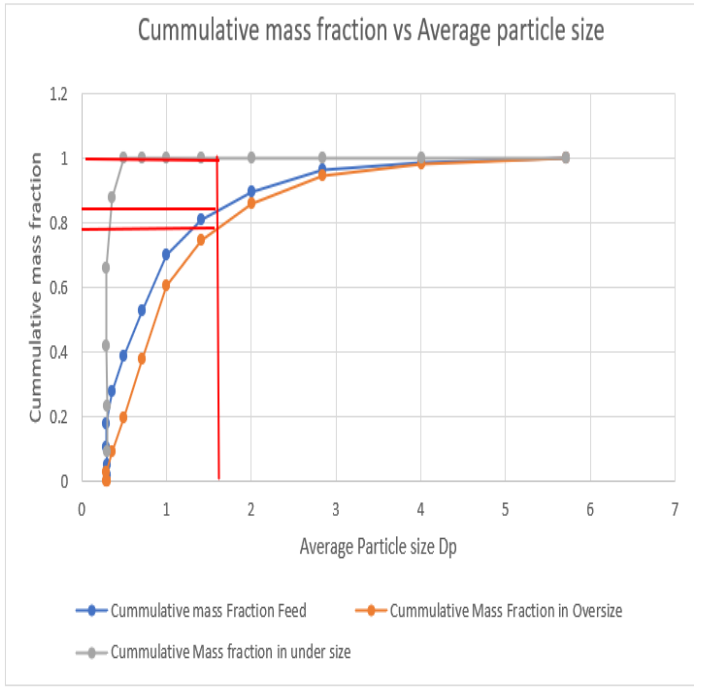
$$\frac{C}{A} = \frac{x_A - x_B}{x_C - x_B} = 1 - \frac{B}{A}$$

Lab Pblm: The following data was obtained from a screen analysis of quartz sample. Find the effectiveness of the screen having average aperture dia 1.6 mm

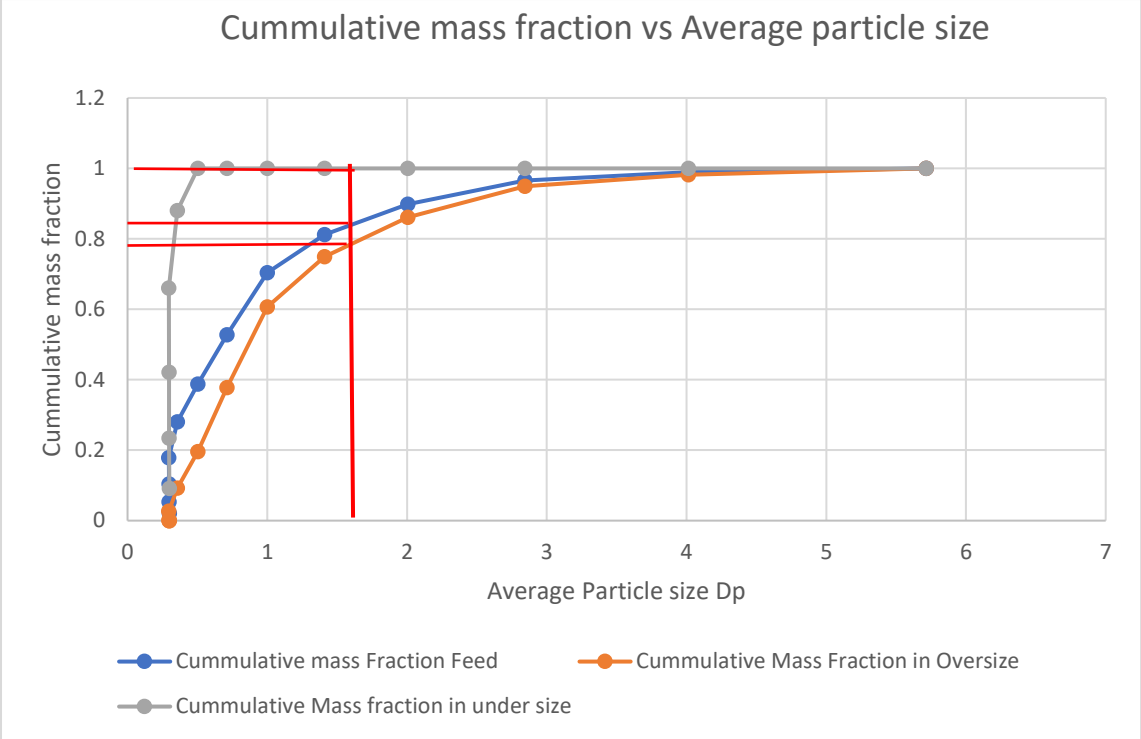
Mesh number	Mesh Number	Aperture Dia (mm)	Mass Fraction Feed	Mass Fraction in Oversize	Mass fraction in under size
	3	6.73			
-3+4	4	4.699	0.0107	0.018	0
-4+6	6	3.327	0.0235	0.033	0
-6+8	8	2.362	0.0672	0.088	0
-8+10	10	1.651	0.0864	0.112	0
-10+14	14	1.168	0.1087	0.142	0
-14+20	20	0.833	0.1759	0.229	0
-20+28	28	0.589	0.1397	0.182	0
-28+35	35	0.417	0.1077	0.104	0.1195
-35+48	48	0.295	0.1013	0.065	0.2198
-48+65	65	0.296	0.0746	0.025	0.2391
-65+100	100	0.297	0.051	0.002	0.1877
-100+150	150	0.298	0.033	0	0.1427
-150+200	200	0.299			

Solution: Calculate average aperture dia and cumulative fractions of feed,oversize and under size. Plot the graph of cumulative mass fraction vs average particle size.

Mesh number	Mesh Number		Mass Fraction Feed	Mass Fraction in Oversize	Mass fraction in under size	Average Dia $\overline{D}_{Pi}$	Cumulative mass Fraction Feed x_A	Cumulative Mass Fraction in Oversize x_B	Cumulative Mass fraction in under size x_C
	3	6.73							
-3+4	4	4.699	0.0107	0.018	0	5.7145	1	1	1
-4+6	6	3.327	0.0235	0.033	0	4.013	0.9893	0.982	1
-6+8	8	2.362	0.0672	0.088	0	2.8445	0.9658	0.949	1
-8+10	10	1.651	0.0864	0.112	0	2.0065	0.8986	0.861	1
-10+14	14	1.168	0.1087	0.142	0	1.4095	0.8122	0.749	1
-14+20	20	0.833	0.1759	0.229	0	1.0005	0.7035	0.607	1
-20+28	28	0.589	0.1397	0.182	0	0.711	0.5276	0.378	1
-28+35	35	0.417	0.1077	0.104	0.1195	0.503	0.3879	0.196	1
-35+48	48	0.295	0.1013	0.065	0.2198	0.356	0.2802	0.092	0.8805
-48+65	65	0.296	0.0746	0.025	0.2391	0.2955	0.1789	0.027	0.6607
-65+100	100	0.297	0.051	0.002	0.1877	0.2965	0.1043	0.002	0.4216
-100+150	150	0.298	0.033	0	0.1427	0.2975	0.0533	0	0.2339
-150+200	200	0.299				0.2985	0.0203	0	0.0912



To find effectiveness of screen having average aperture size 1.6 mm, In the graph mark 1.6 mm on x axis. Drop a vertical line at 1.6 mm, Note down x_A, x_B, x_C. Calculate effectiveness using the formula for effectiveness



Capacity of Screens:

It is measured by the mass of material that can be fed per unit time to a unit area of the screen.

Capacity and effectiveness are opposing factors. To obtain maximum effectiveness, the capacity must be small, and large capacity is obtained only at the expense of a reduction in effectiveness. In practice, a reasonable balance between capacity and effectiveness is desired.

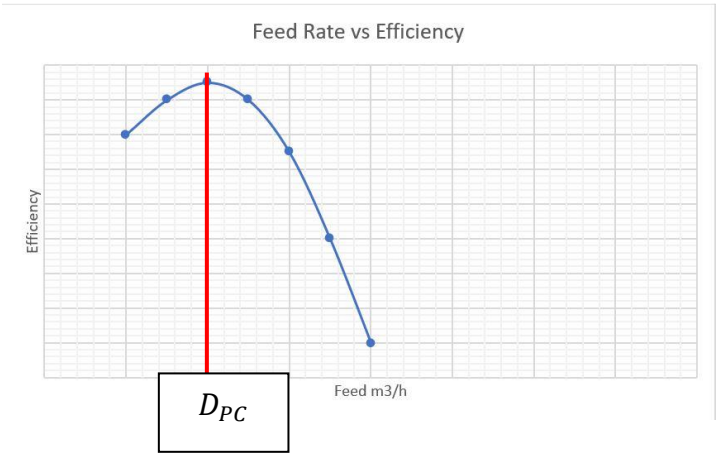
The capacity of a screen is controlled simply by varying the rate of feed to the unit. The effectiveness obtained for a given capacity depends on the nature of screening operation.

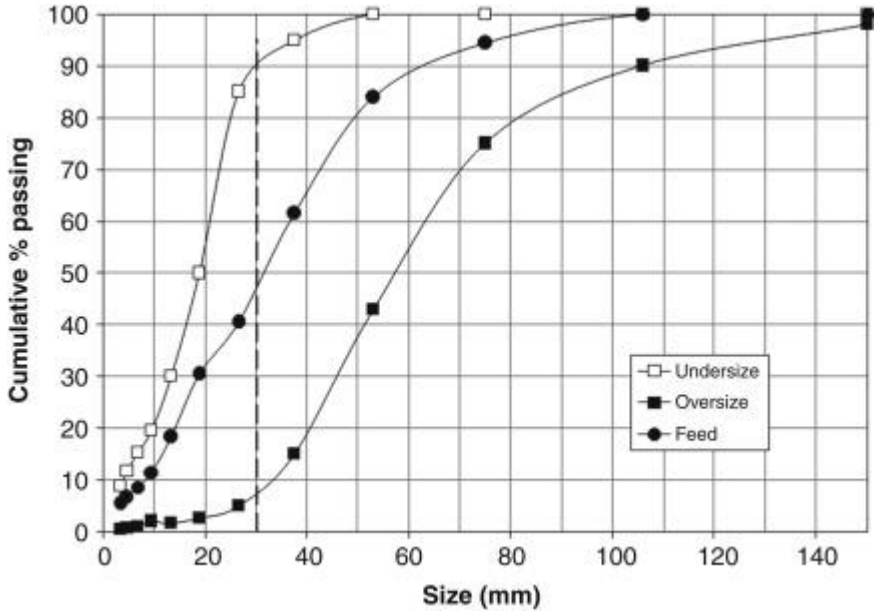
Capacity of Actual Screens:

In practice several complicating factors appear that cannot be treated theoretically. Some of the disturbing factors are the interference of the particles with the motion of any one; blinding, cohesion of particles to each other; the adhesion of particles to the screen surface and the oblique direction of approaches of the particles to the surface when large and small particles are present. The large particles tend to segregate next to the screen and so prevent the smaller particles from reaching the screen. All these factors tend to reduce capacity and lower effectiveness. Moisture content of the feed is especially important. Either dry particles or particles moving in a stream water screen more readily than damp particles which are prone to stick to the screen surface and to each other . As the particle size is reduced, screening becomes more difficult, and the capacity and effectiveness are in general, low for particle sizes smaller than about 150 mesh.

Effect of mesh size on capacity of screens

The probability of passage of particle through a screen depends on the fraction of the total surface represented by openings, on the ratio of the diameter of the particle to the width of an opening in the screen and on the number of contacts between the particle and the screen surface. When these factor are all constant, the average number of particles passing through a single screen opening in unit time is necessarily constant, independent of the size of the screen opening. If the size of the largest particle that can just pass through a screen is taken equal to the width of a screen opening both dimensions may be represented by D_{Pc}.





Experiential Learning

Pb. 100 tons/h of municipality solid waste with composition is shown in Table below is applied to a rotary screen for the removal of glass prior to shredding. Determine the recovery efficiency and effectiveness of the screen based on the following experiment data:

Weight of Under flow = 10 tons/h

Weight of glass in screen under flow = 7.2 tons/h

Component	Mass %	% Moisture content
Food Waste	16.0	60
Paper	40.0	6
Cardboard	8.0	5
Plastics	12.0	2
Textiles	7.0	10
Rubber	4.50	2
Leather	4.50	9
Glass	8.000	0.5

Solution:

$$Effectivensee\ E = Recovery\ x\ Rejection = \frac{B \times x_B}{A \times x_A}\left(1 - \frac{B(1-x_B)}{A(1-x_A)}\right)$$

Industrial Screening Equipments:

Various types of screening equipments have been developed, differing largely in ruggedness, method of moving of the material across them, and materials of construction. A classification based largely on size of material and as follows.

1. Grizzies
2. Trommels
3. Vibrating Screens
4. Gyratory Screen
5. Banana screen

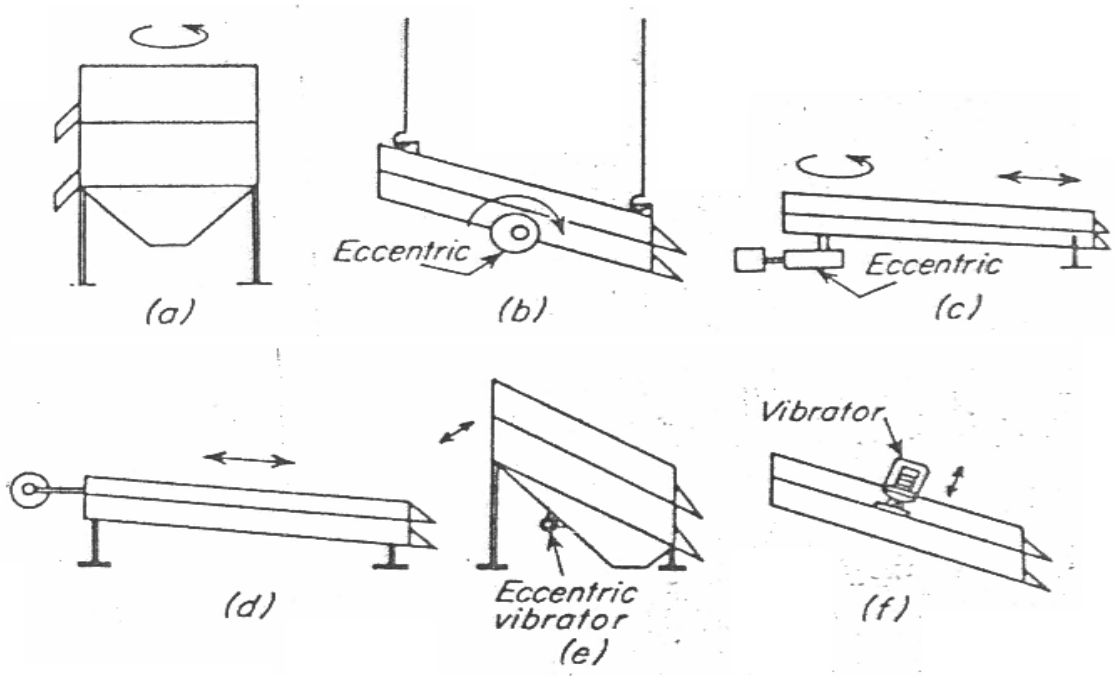
1.

Operating conditions of different screens

Screen	Particle size	Efficiency	Capacity
Grizzlies	20mm – 300mm	Low	Very high
Vibrating screens	480 mesh – 4 mesh	High	High
Gyratory	6mm – 40 mesh	Very high	Low
Trommels	6mm – 55 mm	Average	Low
Banana screen	≈ 100 mm	High	High

1. Screening Equipment:

Many varieties of screens are available for different purposes and only a few representative types are used here. In most screens the particles drop through the openings by gravity. In few designs they are pushed through the screen by a brush using centrifugal force. If fine particles are more the screen surface must be agitated insome way such as by shaking, gyrating or unbarring it mechanically or electrically. Typical screen motions are illustrated in fig.


Eccentric
Eccentric
Vibrator
Eccentric vibrator
(a)
(b)
(c)
(d)
(e)
(f)

1. Grizzlies:


GRIZZLIES
MEKA
Stationary Grizzly

Construction and Working:
A grizzly is a grid of parallel metal bars set in an inclined stationary frame, with a slope of 30°- 45°. The slope and the path of the material are usually parallel to the length of the bars. The length of the bar may be up to 3 meters and the spacing between the bars is 50 - 200 mm.
The material of construction of the bars is manganese steel to reduce wear. Usually, the bar is shaped in such a way that its top is wider than the bottom, and hence the bars can be made fairly deep for strength without being choked by lumps passing partway through them.
Note: A stationary grizzly does not require any power supply and is usually adopted for dry material and is not satisfactory for moist and sticky material. If the grizzly is used for wet screening, a very much smaller angle is employed.

A coarse feed is fed at the upper end of the grizzly. Large chunks roll and side to the lower end while small lumps having a size less than the opening in the bars fall through the grid into a separate collector.

If the angle of inclination to the horizontal is greater, greater will be the output but lower is the screening efficiency. Stationary inclined woven-metal screens operate in the same way that separate particles 12 - 100 mm in size.

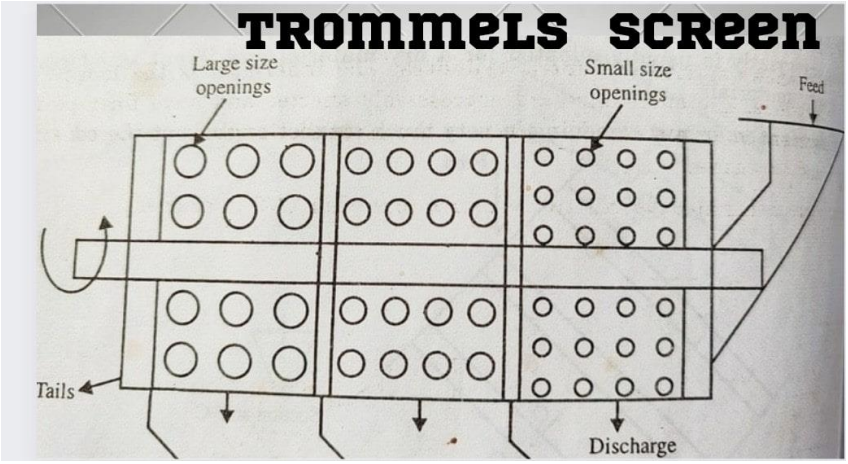
A grizzly finds its greatest application in the separation of the undersize from the feed to a primary crusher. Stationary grizzly is the simplest of all separating devices. It requires no power and is the least expensive to install and maintain. As the openings in the grizzly have a tendency to get blocked by wedge-shaped particles, the labor requirement for operating the grizzly is high. Also, it is difficult to change the opening in the bars.

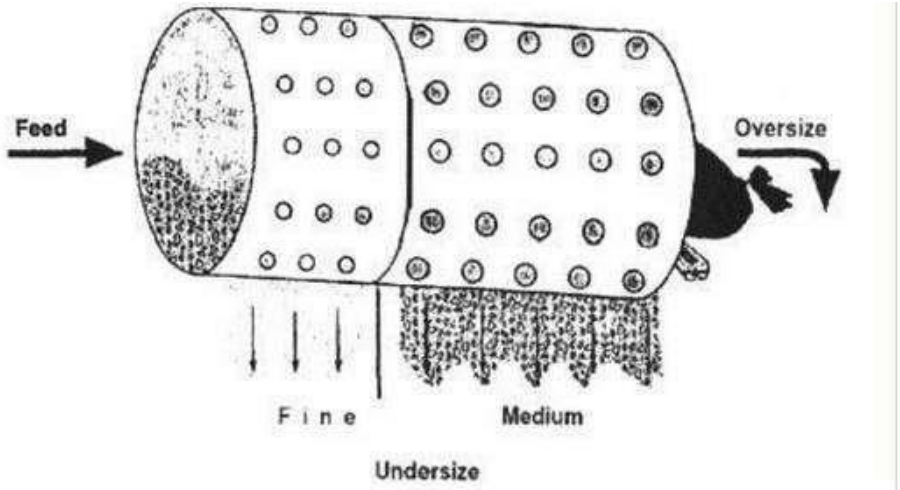
The Grizzly Screen's primary function is to remove oversized material, such as rocks or debris, before it reaches downstream equipment like crushers or screens. This helps protect the downstream machinery from damage and ensures the efficient processing of finer materials. Grizzly Screens are commonly used in industries such as mining, quarrying, construction, and recycling.

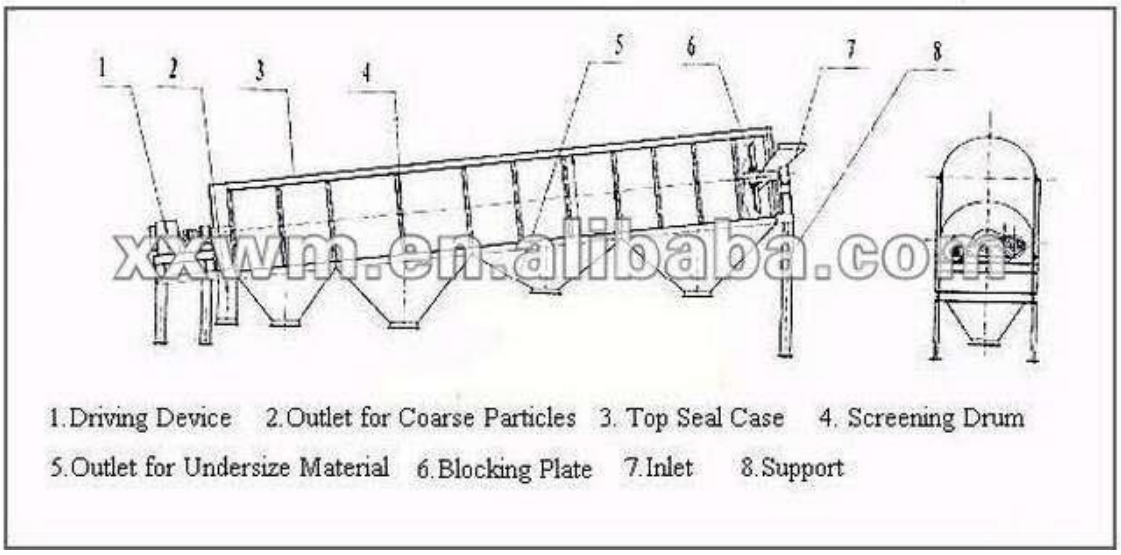
Grizzly Screen Advantages and Disadvantages

These are stationary inclined screens.
Usually, screens are a grid of metal bar which cannot be replaced easly.
The opening of the screen is large.
The handle is large-size feed.
Capacity is large.
Labour requirement is large.
Cheap construction.

TROMMELS
A **trommel screen**, also known as a rotary screen, is a mechanical screening machine used to separate materials, mainly in the mineral and solid-waste processing industries. It consists of a perforated cylindrical drum that is normally elevated at an angle at the feed end. Physical size separation is achieved as the feed material spirals down the rotating drum, where the undersized material smaller than the screen apertures passes through the screen, while the oversized material exits at the other end of the drum.







Construction: Trommels are revolving screens consisting of a cylindrical frame surrounded by wire cloth or perforated plate (which acts as a screening surface). They are open at one or both ends and inclined at a slight angle to the horizontal so that the material is advanced by the rotation of the cylinder. These units revolve at relatively low speeds of 15 to 20 rpm.

A trommel is a mechanically operated screen consisting of a slowly rotating perforated cylinder with its axis at a slight angle to the horizontal.

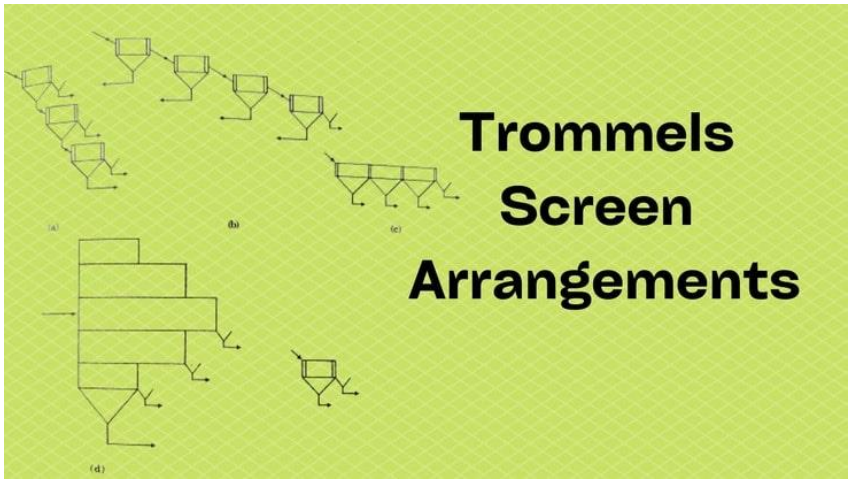
The perforations in the screening surface may be of the same size throughout or may be of different size in which case the small size perforations section is near the feed end. It is driven at the feed end through a gear mechanism. It has a feed point at the upper end, another-sized product discharge below the screening surface, and an oversize discharge at the opposite end.

Trommels Screen Working Principle

The material to be screened is fed at the upper end and gradually moves down the screening surface towards the lower end. In doing so, the material passes over the apertures of gradually increasing size (as the single cylinder is provided with perforations ranging from the finest desired at the feed end to the coarsest at the discharge). If the single cylinder is provided with the screen having three different size perforations then we get four fractions.
The finest material (coarsest) is withdrawn from the discharge end. Such a type of arrangement is usually adopted for smaller capacities. With this type of trommel, there is a tendency of blockage of the apertures by the large material, and the screen with the finest opening being the weakest is subjected to the heaviest wear. **The operating speed of the trommel is 30 to 50% of the critical speed** (critical speed is the one at which the material is carried completely around in contact with the screening surface).

Trommels Screen Arrangements

For the separation of a given material into several size fractions, several trommels are operated in series. The first trommel of a series may have the coarsest perforations so that it produces the coarsest finished product which is delivered to the next trommel and so on. In such a case, it is most convenient to place the trommels one above the other as shown in Fig



When the first trommel of the series has the smallest perforations, the oversize material passes to the next trommel, and so on. In such a case, it is most convenient to put the screens in line, end as shown in fig.

If screening equipment of this type is put into service for finer separations then the cylinder may be covered with a fine wire or silk cloth instead of a perforated plate or coarse wire screen. Such an equivalent is usually called a reel.
Fig. shows several concentric cylinders. The innermost is the longest and has the coarsest perforations. The outer ones are successively shorter and have finer perforations. In this arrangement, the maximum load is given to the strongest screen but the construction is complicated and expensive.
Note: Trommels are well suited for relatively coarse materials (1/2 in or over).

Trommels Advantages and Disadvantage

Advantages

1. These are revolving screens.
2. Labour requirement is low.
3. Opening in small size feed.

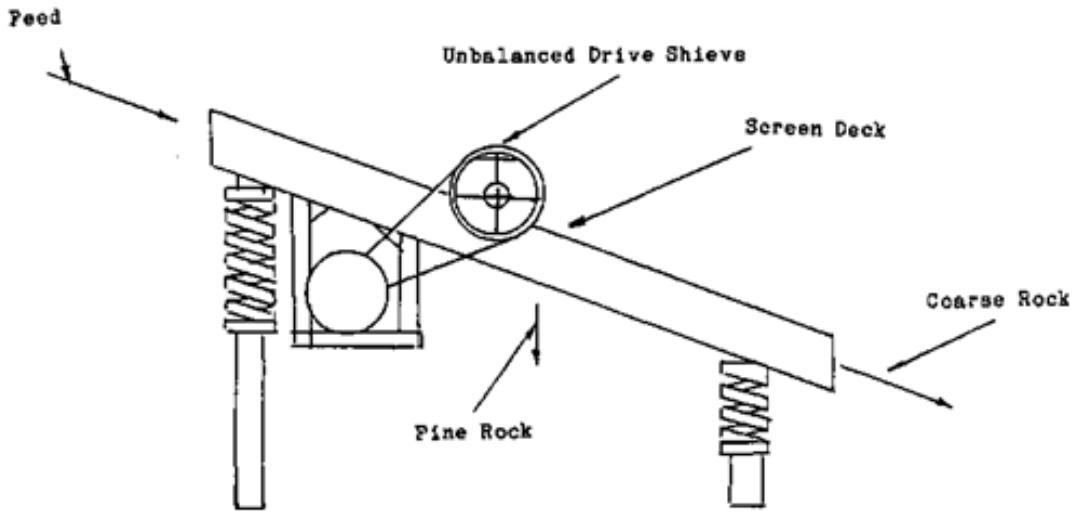
Disadvantages

1. High energy consumption
2. Screen plugging
3. Screen binding
4. Drum damage

Vibrating Screens:
Screens that are rapidly vibrated with small amplitude, keep the material moving and prevent binding as far as possible. **Vibrating screens** are commonly used in industries where large capacity and high efficiency are desired and are able to separate particles between 480 to 4 mesh.







The capacity of the **vibrating screens**, especially in the finer sizes, is so high as compared to the other screens that they have practically replaced all other types where the efficiency of the other screen is of prime importance. These screens are classified as mechanically vibrated screens and electrically Vibrating screens.

The vibrations can be produced either mechanically or Electrically with the frequency of 1800 - 3600 or even more per minute. Mechanical vibrations are usually transmitted from high speeds eccentric to the casing and from the Electrical vibrations are usually transmitted from heavy-duty solenoids directly to the screens so that only screens are vibrated. Vibrating screens may be mounted in a multideck fashion with the coarsest screen at the top, either horizontally or inclined up to 45°.
A directly vibrated screen. Electrically Vibrated screens are widely used in the chemical industry. The Vibrating screens have accuracy of sizing, increased capacity per square metre and low maintenance cost per ton of material handled.

Principle and Operation of the Vibrating Screen

In the case of a vibrating screen, the vibrations are given to the screen to effect the separation of solid particles into two size fractions. Vibrations given keep the solid particles moving. Proper selected frequency and amplitude keep the capacity constant.
In the case of a vibrating screen, the material to be separated is fed to the top screen and simultaneously the screens are vibrated either electrically or mechanically at a frequency of 1000 to 3500 per minute. Due to vibrations, the particles on the screen are kept moving and due to the inclination given to the screens, the oversize material travels along the screen and is collected separately. The undersized material passes through the screens and is collected. Four fractions are obtained with a three-deck screen.

Variables in Screening Operations

These are many variables in a screening operation that can be very easily changed. The best combination of these results in the excellent performance of screening equipment.

Method of Feeding:
In order to obtain maximum capacity and efficiency, the screening equipment must be fed properly. The material should be spared evenly over the full width of the screening surface and should approach the screening surface in a direction parallel to the longitudinal axis of the screen. It must be fed at as low a particle velocity as is possible.Screening Surfaces

Use of single-deck screens in series results in the most efficient operation as in the case of multiple-deck screens lower decks are not fed so that their entire area is not used and each separation requires a different combination of angle, speed, and amplitude of vibration for the best performance.

Screen Slope

As the screen slope increases, the rate at which the materials travel over the screening surface increases, and at the same time, it reduces the bed thickness. The increase in the rate of travel means an increase in tonnages passing over the screen per unit of time.
Reduction in the bed thickness allows the fines to work their way to the screen surface and pass through the openings. However, the slope cannot be increased beyond a certain value because beyond that value the material will travel down the screen much faster without getting screened and the screening efficiency reduces drastically.

Vibration Amplitude and Frequency

One has to select the proper amplitude of the vibrations to prevent binding of the cloth and for long bearings life. The frequency of vibration affects the capacity of the screening equipment by regulating the number of impacts between the material and the Screening surface. While selecting both these parameters, one has to see that the feed is properly stratified for the most efficient operation.

Moisture in Feed

The moisture associated with the feed material adversely affects the screening operation and should be removed.

Vibrating screen Advantages and Disadvantages

Advantages

1. To a strong vibration of the screens box, so that the vibration screen is high efficiency and productive.
2. Dry material screening to the demand, and lightweight, more layer.
3. Easy to use and replace the screen surface.
4. Less consumption of electricity per ton of material.

Disadvantages

1. Not suitable for high moisture.
2. Vibration Screen used for large scale and the model is relatively high energy consumption.
3. Work noise and dust are heavier.

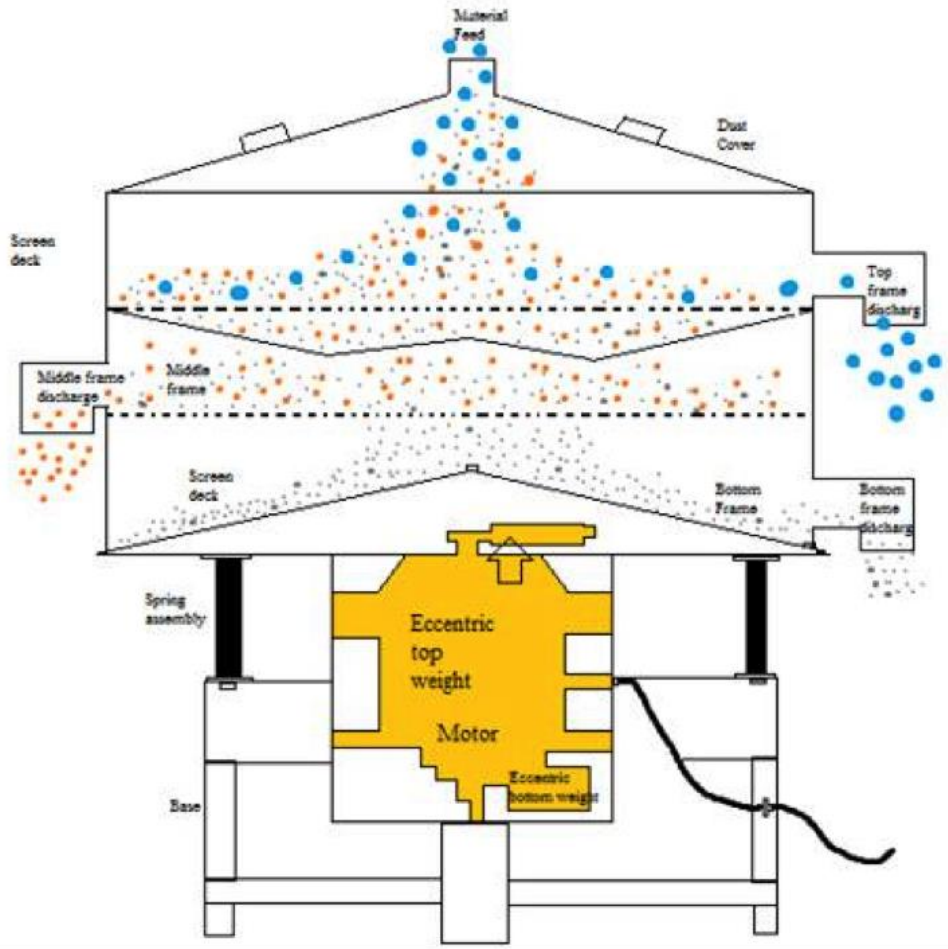
Gyratory Screens: Gyratory equipment, used in mechanical screening and sieving is based on a circular motion of the machine. Unlike other methods, gyratory screen operates in a gentler manner and is more suited to handle fragile things, enabling it to produce finer products. This method is applicable for both wet and dry screening.

The gyratory equipment contains decks of cards on top of each other with the coarsest screen on top and the finest below. The feed is inserted from the top and gyratory motion triggers the penetration of particles into the next deck through screen openings.

Casings are inclined at relatively low angles (< 15°) to the horizontal plane, with gyrations occurring in the vertical plane. The eccentric masses can be varied in such as the increase of top eccentric mass leads to an increase in horizontal throw, promoting the discharge of oversize materials. Increment in bottom eccentric mass boosts the material turn over on the screen surface, maximizing the quantity of undersize-material penetration. Oversize materials are discharged via tangential outlet.

The option to select number of decks enables gyratory equipment to accurately separate materials consisting particles that are very close in size. This advantage is unrivalled and proves to be significant in the powder processing industry where fine materials are involved. High separating efficiency and ease of maintenance puts gyratory screening ahead compared to other processes in terms of product quality.

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Industry	Applications
Process	Processing ceramics, pulp and paper mill, paints, sand, starch slurry
Food	Screening of refined table salt, papaya cubes, turmeric pigment; clarification of alkaline extracts
Chemical	Screening hydrate lime, effluent overflow from hydrocyclone; classification of polyester beads, anhydrous aluminium chloride

Advantages

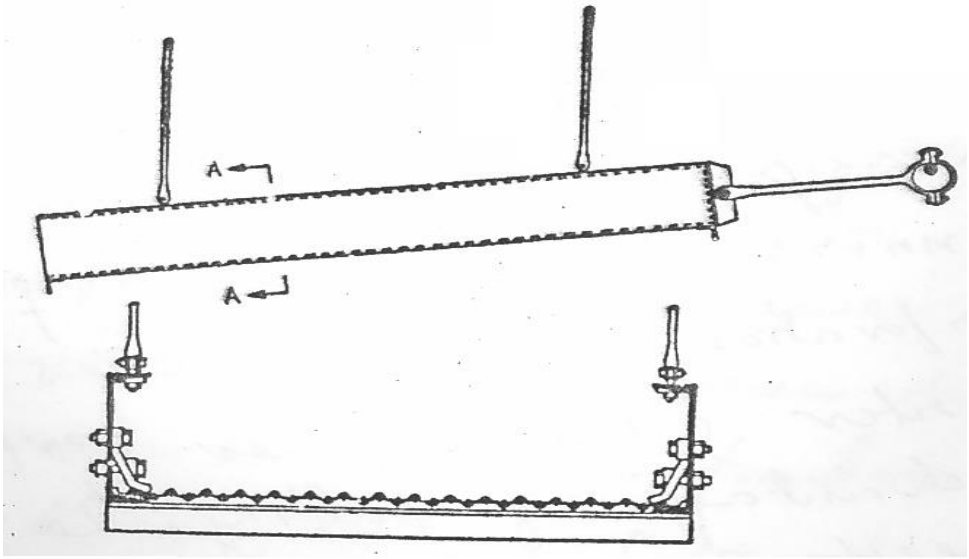
- Low running cost
- Ideal for multi-fraction separation
- Flexible range of applications
- Good efficiency and quality of separation
- Easily maintained
- Low screen blinding

Limitations

- Large amount of floor space
- Relatively difficult to operate
- Susceptible to lumps and agglomerates in the feed

3. Shaking Screens:

Many size separations in which the product may be from ½ inch down to most finest sizes that can be handled by screens, may be performed by means of flat or slightly inclined screens that are given a reciprocating motion. Fig. Below shows such a screen made of single mechanical elements.



The frame is of channel irons and is suspended by layer rods so that it can move freely. It is shaken by means of an ordinary eccentric on a rotating shaft. The screen cloth may be riveted directly to the frame, or it may be soldered. Over a light, removable frame bolted into place. Another method of attachment is to provided a light angle that can be bolted to the inside of the frame. The edge of the screen cloth is drawn up between this angle and the man structure.

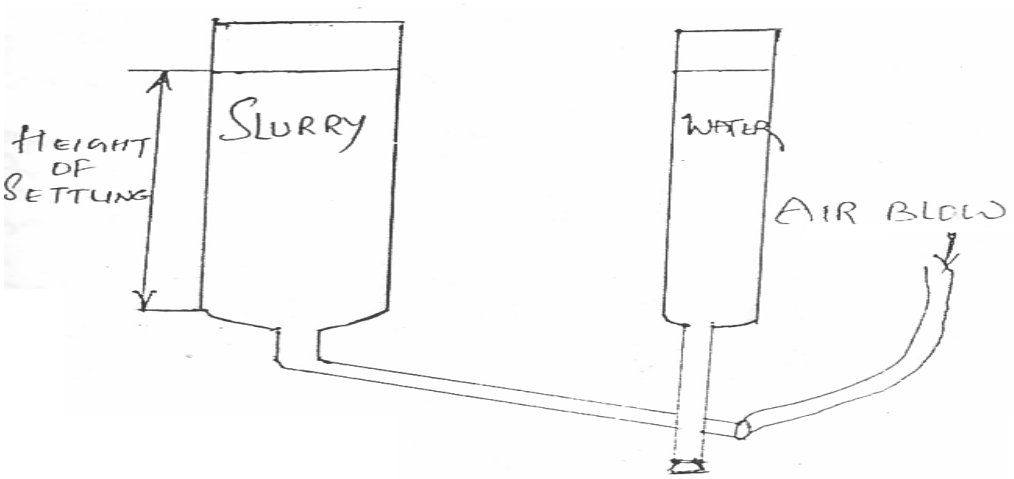
Sub-Sieve Analysis:

The analysis are done using settling techniques. The methods used in general are sedimentation. Air permeating method and Elutriation methods.

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ICI Sedimentation:



This is a method developed by imperial chemical industries, U.K for the size distribution of a powder sample and to calculate the average size. In this method different sizes of particles are assumed to be present and the settling velocities are calculated assuming free settling under stokes region. The time of settling of different particles are calculated using settling velocities and the mixture of particles are separated into different fraction by allowing the particles to settle for different calculated times.

The settling velocities are calculated using the equation $u_t = \frac{gD_P^2(\int_P - \int)}{18\mu}$

Time of Settling = $\frac{height\ of\ settling}{Ter\,min\,al\ settling\ velocity}$

Weigh a known quantity of sample . Prepare thick slurry of the sample with water and take it in the left limb (Settling limb) of the sedimentation apparatus. Add water to the right limb up to some known height and make up the slurry in the left limb and equalize the level in the limbs to the known height. To keep the slurry agitated and keep it uniform blow air through the rubber tube provided at the bottom. Assume that the particles present area of sizes 60μ, 60/√2 μ, 30 μ, 30 / √2μ and calculate the times of settling for each of them. After making the slurry uniform stop blowing of air, start the stop watch and allow the solids to settle for the calculated time of 60 μ particles. At this moment, close the stop cock in the left limb. Open the outlet at the bottom of the left limb and also open the stop cock in the right limb to flush the settled solids out. Filter the slurry in a previously weighed filter paper, wash the solids present, dry and weigh. Repeat the procedure for the other sizes also and tabulate the values as rough weights W_1, W_2, W_3 and W_4. Calculate the connected weights using the formula

$X_1 = W_1 - W_2$, $X_2 = 2\,W_2 - W_3$, $W_3 = 2W_3 - W_2$

$W_4 = \text{Total} - X_1 - X_2 - X_3$.

Calculate the average size of the particles by differential or cumulative methods of analysis and report.

Air Permeability:

The test is intended for the determination of the specific surface area of dry powders expressed in square metres per gram in the sub-sieve region (<30 µm particle size).
The **air permeability specific surface** of a powder material is a single-parameter measurement of the fineness of the powder. The specific surface is derived from the resistance to flow of air (or some other gas) through a porous bed of the powder. The SI units are $m^2 \cdot kg^{-1}$ ("mass specific surface") or $m^2 \cdot m^{-3}$ ("volume specific surface"). Air permeability is almost universally used in the cement industry as a gauge of product fineness which is directly related to such properties as speed of setting and rate of strength development.

Other fields where air permeability has been used to determine specific surface area include:

- Paint and pigments
- Pharmaceuticals
- Metallurgical powders, including sintered metal filters.

Measurement consists of packing the powder into a cylindrical "bed" having a known porosity (i.e. volume of air-space between particles divided by total bed volume). A pressure drop is set up along the length of the bed cylinder. The resulting flow-rate of air through the bed yields the specific surface by the Kozeny–Carman equation:

$$Sw = \left[\frac{\epsilon^3 \times \Delta P \times 1}{5(1-\epsilon)^2 \, L \rho_s^2 \, v\mu}\right]^{1/2} m^2/kg$$

ε = Porosity of bed = $\frac{Void\ Volume}{Bed\ Volume}$ $= \frac{1/\int_b - 1/\int_P}{1/\int_b} = \frac{mass\ of\ sample}{volume\ of\ bed}$, kg/ m^3.

$$\int_B = density\ of\ Bed \qquad \int_P = density\ of\ particles\ or\ solid$$

Volume of bed = C/S area of bed × height of bed, m^3

µ = Viscosity of fluid (Air) = 18.325×10^{-6} NS / m^2.

$\int_s = density\ of\ particles\ or\ solid$

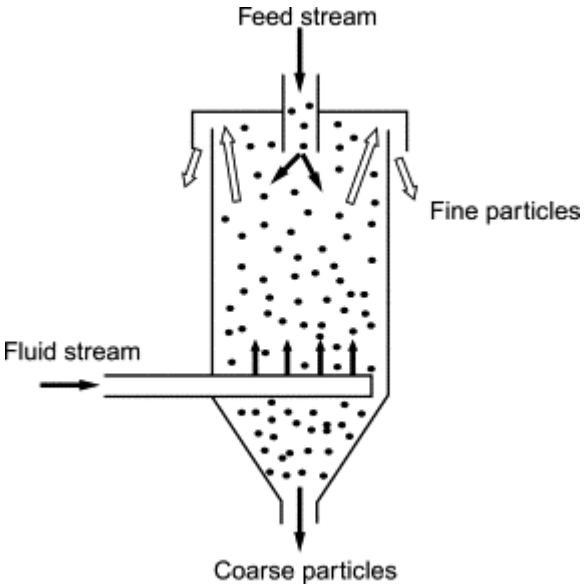
V_O = Superficial velocity of fluid flowing (air) m/s.

$$= \frac{volume\ of\ water\ collected}{Time\ of\ collections\ C/S\ Area\ of\ \ bed}$$

L – Height of bed, m,
$\Delta P = R_m \int g$, Pressure drop for the flow of fluid, N/m^2.
R_m – manometer reading
$\int$ - density of manometric fluid = kg/m^3.
G – Acceleration due to gravity. 9.81 m/s^2.

Air Elutriation

Elutriation is a process for separating particles based on their size, shape and density, using a stream of gas or liquid flowing in a direction usually opposite to the direction of sedimentation. This method is mainly used for particles smaller than 1 μm



Elutriation is also used in the centrifugal field, usually with air as the fluid. In principle, air with entrained particles travels inward against the centrifugal force. Particles with sufficiently low terminal velocities also pass inward; the rest pass outward.

Elutriators use the velocity of an ascending stream of fluid to affect separation. The feed mixture containing solid particles of different sizes is fed from the top and the fluid (normally water or air) is sent up from the bottom. Particles whose free settling velocity is more than the upward velocity of fluid settle down and are collected as the underflow from the bottom of the chamber, whereas particles whose free sterling is less than the upward velocity of fluid air carried over by the ascending fluid are carried over and are collected in the over flow at the top of the column. By using a number of elutriation column of increasing cross-sectional area in series, several fractions of solids can be sorted.